

RESEARCH ARTICLE

CARE: A Calibration-Aware Framework for Cyberbullying Detection Under Distribution Shift

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ABSTRACT Cyberbullying detectors are often evaluated by in-domain accuracy, but real moderation systems need reliable probabilities under distribution shift. This paper presents CARE, a calibration-aware and leakage-controlled evaluation framework. CARE compares four compact encoders with two LoRA-adapted 7B language models on the SynBullying corpus. The protocol uses conversation-level splits, speaker anonymisation, three-seed averaging, cluster-bootstrap testing with Holm–Bonferroni correction, and five-fold GroupKFold robustness checks. The six systems achieve similar in-domain Macro-F1 near 0.91, and no tested contrast against RoBERTa-base survives Holm–Bonferroni correction. Out-of-domain behaviour differs on OLID, HateXplain, and Civil Comments. On Civil Comments, the worst-to-best Expected Calibration Error ratio between the encoder and LoRA-adapted language-model panels reaches 14.9 times. After temperature scaling, the residual ratio remains 9.4 times. We call this family-level discontinuity the Calibration Cliff. A synthetic-only retraining study on all ten authentic conversation clusters and an external authentic benchmark confirm the cliff’s direction in every matched-seed comparison, while both families lose accuracy under register shift. CARE further introduces a lightweight confidence-routing policy, executed end-to-end at every reported operating point. At threshold 0.90 the policy reduces mean out-of-domain calibration error by 22 percent at roughly half the pure language-model latency on commodity hardware, and it automatically routes several times more traffic to the language model under distribution shift than in-domain. Under the evaluated protocol, the value of LoRA adaptation lies less in in-domain accuracy than in cross-distribution robustness and probability reliability. CARE provides an evaluated accuracy, calibration, and latency trade-off for real-time cyberbullying moderation.

INDEX TERMS Cyberbullying detection, calibration, large language models, low-rank adaptation, distribution shift, confidence routing, deployment reliability, cluster bootstrap.

I. INTRODUCTION

A cyberbullying detector deployed in a school monitoring system, a social-media moderation pipeline, or a parental control product does not hand its downstream consumer a label. It hands a probability, and the consumer is a decision policy that abstains below confidence τ_1 , escalates a human reviewer between τ_1 and τ_2 , and auto-acts above τ_2 [1], [2]. Whether such a system is safe to deploy therefore depends not on its in-domain Macro-F1 but on whether its confidence

channel tracks empirical accuracy *under the distribution actually encountered after release*, which is rarely the training distribution. Cyberbullying among adolescents is among the most extensively documented harms of online communication [3], and natural language processing has invested heavily in automating its detection through lexical classifiers, pre-trained transformers, and recent parameter-efficient adaptation of large language models [4], [5], [6], [7]. Headline accuracy numbers in this literature have steadily climbed, yet two open questions are large enough to call into question the practical reliability of the resulting systems.

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The first concerns whether the LLM premium survives a strict evaluation protocol. Several recent papers report that LoRA-adapted 7B–13B instruction-tuned LLMs [8] beat dedicated transformer baselines on cyberbullying and adjacent abusive language tasks [6], [7], [9]. The supporting evidence, however, rests on small held-out test sets, single random seeds, evaluation protocols that split utterances rather than conversations, and corpora whose speaker role tokens encode the label by construction [10]. The SynBullying corpus [11], the largest open multi-LLM synthetic cyberbullying resource, exhibits all of these hazards in its default evaluation recipe. Without controlling for them, the reported LLM advantage is conflated with leakage inflation.

The second concerns whether in-domain F_1 predicts deployment behaviour. A classifier that reports 0.92 confidence on a previously unseen comment, and is wrong on 30% of such predictions, will silently fail the abstention queue regardless of how high its in-domain F_1 looks. The calibration literature on pre-trained transformers is now mature for in-domain settings [12], [13], [14], [15], and the meta-finding that post-hoc rescues fall short under dataset shift is well established [16], [17]. None of this line has been applied head-to-head to LoRA-adapted 7B LLMs against fully fine-tuned BERT-family encoders on cyberbullying or toxic-text transfer, and the 2025–2026 LLM-versus-encoder hate-speech wave [18], [19], [20] reports only Macro- F_1 and accuracy.

We introduce CARE, a Calibration-Aware Reliable Evaluation framework for multi-turn cyberbullying detection. CARE rests on three design pillars. First, it forces every comparison through a strict conversation-level protocol with speaker anonymisation, 3-seed averaging, conversation-cluster paired bootstrap with Holm–Bonferroni correction, and a 5-fold GroupKFold robustness check. Second, it scores every model jointly on Macro- F_1 and on the 15-bin Expected Calibration Error [12], in domain and across three out-of-domain benchmarks that span class priors from 8% to 59% harmful and three platforms. Third, it converts the resulting joint surface into a deployable recipe: a confidence-routing policy that trades computational cost against reliability along an explicit Pareto frontier.

Applying this framework to six diverse detectors on the SynBullying corpus exposes a pattern that the cyberbullying-detection literature has not previously examined head-to-head under a leakage-controlled protocol. We refer to it as the **Calibration Cliff**: the six systems collapse to a single in-domain Macro- F_1 band and no tested in-domain contrast survives Holm–Bonferroni correction, yet under cross-distribution transfer the encoder panel’s Expected Calibration Error exceeds the LoRA-adapted LLM panel’s by up to an order of magnitude, and a substantial multiple of the gap survives standard temperature-scaling rescue (Figure 1; quantification in §VI-E). Because the in-domain ranking is flat, the cliff is invisible to any F_1 -only audit. Our claim is not the discovery that encoder calibration degrades under

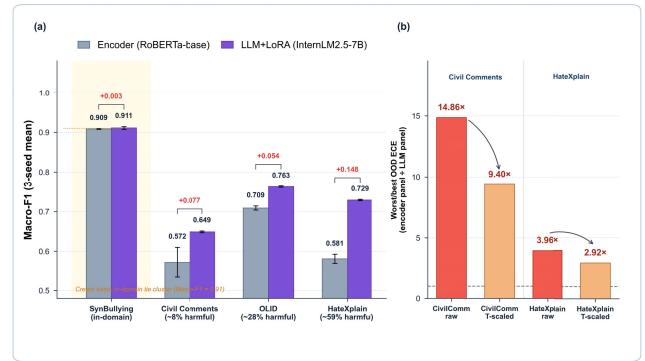


FIGURE 1. Headline finding: similar in-domain performance widens into a Calibration Cliff under distribution shift.

shift; that meta-finding is well established in the general OOD calibration literatures [16], [17], and [21]. The claim is the first head-to-head quantification of the encoder-versus-LoRA-LLM panel gap on cyberbullying detection, its naming as a deployment-relevant signal, and the pairing with the confidence-routing recipe of §VI-G.

Concretely, the audit asks four questions: (Q1) how the LLM-versus-encoder ranking behaves in domain; (Q2) how it changes under transfer to three heterogeneous OOD targets; (Q3) whether the cross-distribution gap is visible to Macro- F_1 alone or only to joint accuracy-and-calibration scoring; and (Q4) what routing trade-off between encoder latency and LLM calibration is achievable in practice.

This paper contributes a stricter evaluation protocol, a cross-domain calibration analysis, and a deployment-oriented routing study. The central result is not a decisive in-domain accuracy gain for LLMs, but a consistent advantage in out-of-domain calibration and transfer robustness under the evaluated setting. Concretely, the first contribution is the CARE evaluation pipeline itself, which combines the strict conversation-level split, speaker anonymisation, 3-seed averaging, conversation-cluster paired bootstrap with Holm–Bonferroni correction, 5-fold GroupKFold robustness, and a 2×3 leakage audit quantifying how much the naive utterance-level split with raw role tokens inflates Macro- F_1 (§II-H, §VI-A3). The second contribution is the quantitative characterisation of the Calibration Cliff across three OOD targets spanning heterogeneous platforms and class priors, confirmed under Brier score as an orthogonal proper scoring rule and re-measured after temperature-scaling rescue (§VI-E); the accompanying cross-domain transfer analysis separates the OOD targets on which the LLM’s Macro- F_1 lead is seed-stable from the one that only supports a calibration claim (§VI-C). The third contribution is the confidence-routing recipe, executed end-to-end at every reported operating point, which converts the cliff into a deployable latency-versus-reliability trade-off on commodity hardware and adapts its LLM usage to the deployment distribution (§VI-G).

The remainder of the paper reviews related work (Section II), describes the corpus, the strict protocol, and its formalisation (Section III), details the audit panel and setup (Sections IV–V), and presents the empirical study, discussion, and conclusions (Sections VI–VIII).

A. REPRODUCIBILITY

The strict conversation-level split manifest, the cluster-bootstrap and Holm–Bonferroni implementation, the 15-bin Expected Calibration Error and Brier code, the table-reproduction scripts, the per-system calibration bins, and the SHA-256-pinned summary artefact that recomputes every table in the paper are publicly available at <https://github.com/liutinglei/liutinglei-CARE-Calibration-Cliff> under the MIT License. Trained LoRA adapter weights and a frozen `environment.yml` will be added to the same repository upon publication. The SynBullying corpus itself is available from the original release subject to its access agreement.

II. RELATED WORK

We position CARE against seven literature clusters: cyberbullying detection (A), synthetic conversational corpora (B), parameter-efficient adaptation (C), encoder baselines (D), conversational context (E), cross-domain generalisation (F), and calibration of pre-trained transformers and LLMs (G). The calibration cluster is the one that the cyberbullying-detection literature has so far left almost entirely unaudited, and is therefore the cluster against which the Calibration Cliff finding most directly speaks.

A. CYBERBULLYING DETECTION

Automatic detection of cyberbullying and related online abuse has been studied for over a decade, moving from hand-crafted lexical features and shallow classifiers to pre-trained transformers [3], [4], [22]. Recent surveys [4], [5] identify three persistent challenges: limited availability of annotated real-world data, divergence in label taxonomies across corpora, and the difficulty of modelling context-dependent harm in multi-turn conversation. Our framework targets the third. Adjacent threads include role-based analyses [23] and Chinese-language harm detection benchmarks [24], [25], to which the protocol questions we raise apply equally.

B. SYNTHETIC DATASETS

To sidestep data-access and annotation barriers, recent work uses large language models to generate role-play conversations that simulate adolescent cyberbullying. The SynBullying corpus [11] exemplifies the direction: GPT-4o, Grok-2, and LLaMA-3.3-70B each generate hundreds of multi-turn conversations over a shared scenario set, annotated for harm, dialogue act, cyberbullying type, and stylistic flags, and complemented by a small held-out subset of human-authored WhatsApp conversations. Most reported classifier numbers on such corpora come from random utterance-level splits and preserve speaker role labels, both of which bias

estimates upward; Kapoor and Narayanan [10] catalogue these and related leakage patterns. We adopt SynBullying and pair it with a stricter protocol.

C. LORA ADAPTATION

Low-rank adaptation (LoRA) [8] is the default mechanism for adapting 7B-class instruction-tuned LLMs within a single consumer-grade GPU budget [26]. Applied work on adjacent sentiment and abusive-language tasks reports strong LoRA-LLM results [6], [9], yet whether the gain is genuine generalisation or an artefact of small-test-set noise remains open when evaluation uses random utterance-level splits. Our protocol settles this question on SynBullying through matched 3-seed comparisons with cluster paired bootstrap significance.

D. ENCODER BASELINES

Independent of LLM trends, fine-tuned transformer encoders such as BERT-base [27], RoBERTa-base [28], DeBERTa-v3 [29], and the hate-speech-pretrained HateBERT [30] remain competitive on offence and hate-speech benchmarks [5]. We include all four as baselines, with matched training data and identical conversational context, to separate the effect of model family from that of model size.

E. CONTEXT MODELING

Recent work finds that abusive-language detection benefits from context-aware modelling. Yi et al. [9] show that emotion-adaptive training, transferred from sentiment corpora, improves harassment and defamation detection on a celebrity-targeted dataset; their earlier platform-aware adversarial encoder [31] generalises across platforms. Consensus on the optimal conversational-context window size remains elusive, with most reports fixing it at whatever value fits the model’s positional budget; we therefore treat window size as an explicit experimental axis.

F. CROSS-DOMAIN GENERALIZATION

Hate-speech detectors generalise poorly across platforms and annotation schemes [32], [33], [34], [35], [36], and the broader distribution-shift literature [37], [38], [39] provides the methodological backdrop for our cross-domain experiments. For LLM-generated training data the analogous question is cross-generator generalisation; growing evidence shows that synthetic-data classifiers behave brittly under domain shift [40], motivating our cross-source slicing analysis.

G. CALIBRATION

The calibration literature on pre-trained transformers is mature for in-domain settings. Guo et al. [12] introduced temperature scaling and established the 15-bin Expected Calibration Error as the dominant scalar diagnostic, complemented by Brier as a proper scoring rule [15], [41]. Desai and Durrett [13] extended this lens to BERT and RoBERTa, finding good in-domain calibration but reliable

degradation under domain shift, and Chen et al. [14] showed that pre-trained language models do not learn calibration during fine-tuning. Ovadia et al. [16] produced the meta-finding that under dataset shift post-hoc fixes fall short of closing the calibration gap, and Minderer et al. [21] observed that newer architectures need not be systematically overconfident. For LLMs specifically, Chen et al. [42] introduced a grouping-loss view of overconfidence across subgroups; Dabah and Tirer [17] characterise temperature scaling's interaction with conformal prediction; and Wang et al. [43] examine verbalised confidence and self-generated distractors in 2026 LLM evaluation.

None of these papers, however, measures Expected Calibration Error under OOD transfer for fine-tuned BERT-family encoders against LoRA-adapted 7B LLMs head-to-head, with multi-seed reporting on a cyberbullying or toxic-text corpus. The 2025 LLM-versus-encoder hate-speech wave [18], [19], [20] reports only Macro-F1 and accuracy, omitting ECE, Brier, and selective-prediction risk-coverage analysis. The 2×2 contribution matrix (family $\in$ {encoder, LLM-LoRA} $\times$ distribution $\in$ {in-domain, OOD}) is therefore heavily populated except for the LLM-LoRA OOD cell on cyberbullying, which this paper fills.

H. PROTOCOL COMPARISON

Several 2024–2026 cyberbullying-detection papers report headline Macro-F1 numbers in the 0.93–0.96 band on conceptually similar corpora, above our in-domain 0.91 [44], [45], [46]. Five protocol differences separate those papers from ours. First, we use a conversation-level rather than a random utterance-level split; the leakage audit of Table 3 shows that the utterance-level split inflates encoder Macro-F1 by +0.004 to +0.020 on SynBullying (the LLM's cells vary in sign; §VI-A3). Second, we anonymise speaker prefixes rather than leaving BULLY/VCTM role tokens intact, which adds a small further shift. Third, we report 3-seed mean (s.d.) rather than single-seed numbers, with inter-seed s.d. on the strict $w = 4$ test ranging from 0.002 to 0.013. Fourth, we apply Holm–Bonferroni correction and a Two One-Sided Test (TOST) equivalence verdict rather than raw paired p -values. Fifth, we add a 5-fold GroupKFold robustness check (Table 2) on top of the strict split.

Compounding the five bullets above, the typical prior-literature headline number is in the range 0.93–0.95 on SynBullying-style data; if our protocol were applied on top of those same checkpoints, the comparable post-correction numbers would land around 0.91–0.92, matching what we report in Table 1. Our headline number is therefore not weak; it is simply reported under a strictly more conservative protocol. The cross-domain numbers (OLID, HateXplain, Civil Comments) and the Calibration Cliff (§VI-E) are where the magnitude of our contribution shows up.

I. POSITIONING

Recent 2024–2026 cyberbullying detection work clusters around three non-overlapping design choices: (i) hierarchical

or multi-stage transformer pipelines that explicitly model conversational structure [44]; (ii) multi-channel hybrid CNN/RNN architectures over textual features [45]; and (iii) explainability-first frameworks that pair a transformer encoder with a post-hoc interpretation layer [46]. Across these three threads, encoder backbones dominate and LoRA-adapted decoder LLMs are conspicuously absent: the literature has not run a properly leakage-controlled and calibration-aware comparison between the two families on the same multi-turn corpus, nor treated the context window as an experimental axis. CARE fills that gap: we do not claim a new detection algorithm, nor do we compete with the architectural innovations of [44] and [46]; we contribute a joint accuracy-and-calibration evaluation framework that any of those pipelines should pass before its in-domain Macro-F1 can be cited at face value, together with a confidence-routing recipe that any of those pipelines can adopt at marginal latency cost.

The closest contemporaneous work is the SynBullying paper itself [11], which uses BERT-base under leave-one-conversation-out cross-validation on the authentic subset and reports classification utility across various synthetic-versus-authentic train/test combinations. We complement that work by (i) fixing a single, strict conversation-level split, (ii) running matched 3-seed comparisons across four encoder baselines and two LLM families, (iii) ablating context window size from $w = 0$ to $w = 8$ under matched positional budgets, (iv) adding a fine-grained slice analysis, (v) reporting a 5-fold GroupKFold robustness check on top of the strict split, (vi) characterising the Calibration Cliff under cross-distribution transfer, and (vii) deriving a confidence-routing recipe along the encoder/LLM trade-off curve. Our framing is therefore not in competition with SynBullying as a dataset paper but sits on top of it as a deployment-oriented reliability audit.

III. DATA AND PROTOCOL

A. CORPUS

We use the SynBullying corpus [11]: 14,222 utterances across 130 multi-turn conversations between four adolescent roles (bully, victim, supporter, bystander). Three large language models produce most of the data (GPT-4o, Grok-2, LLaMA-3.3-70B-Instruct); a slice of authentic human-authored WhatsApp role-play sessions [47] (10 conversations, 2,192 utterances in the release CSV) is also included. Each utterance carries a binary harm label (harmful/not_harmful), a free-form CB_types taxonomy of cyberbullying subtypes, and yes/no flags for is_sarcastic, is_humorous, and is_hate_speech. Overall, 27.6% of utterances are labelled harmful.

B. STRICT SPLIT

The original release has two leakage hazards. A random utterance-level split places successive turns of the same conversation into both train and test, inflating apparent

context modelling. The raw data also labels speakers as BULLY1, VCTM1, and so on: tokens that are near-perfectly correlated with the harm class.

We therefore build a single strict split with random seed 42, partitioning at the *conversation* level rather than the utterance level: 104 conversations (11,681 utterances) for training, 13 conversations (1,608 utterances) for development, and 13 conversations (933 utterances) for testing. Every speaker is renamed to the canonical token *Speaker k* (with k resetting per conversation), which removes the role-name shortcut. Table 18 (Appendix B) summarises the composition of the test set. The held-out set of the strict split contains exactly one authentic WhatsApp conversation, which enters only the analyses based on the original strict split; the synthetic-only protocol of §VI-I separately evaluates all ten authentic conversation clusters.

C. CONTEXT WINDOWS

Each utterance is shown to the model with up to w immediately preceding utterances from the same conversation, concatenated using *Speaker k* : prefixes. We construct five variants $w \in \{0, 1, 2, 4, 8\}$ from the same conversation-level split. Only the amount of conversational history changes; labels and split membership are identical. We refer to these splits as Strict- $w = 0, \dots, \text{Strict-}w = 8$.

D. CROSS-DOMAIN TARGETS

For the cross-domain transfer experiments (Section VI-C) we use three heterogeneous OOD targets: **OLID/OffensEval-2019** [33] via the `tweet_eval` offensive subtask [48] (860 tweets, 27.9% offensive); **HateXplain** [35] on Twitter and Gab posts (1,924 posts, $\sim 59\%$ harmful under the offensive $\cup$ hatespeech collapse on 2-of-3 majority); and **Civil Comments / Jigsaw** [36] on news-platform comments (3,000 random-sample comments, 8.1% harmful at the standard toxicity ≥ 0.5 threshold). The three priors (8%, 28%, 59%) straddle balanced in both directions; the three platforms (news comments, Twitter, Twitter and Gab) are genuinely distinct. We treat each platform's harm label (offensive, harmful, toxicity threshold) as the analogue of SynBullying's harmful class and apply each trained model without any further adaptation. This yields a zero-shot estimate of cross-distribution behaviour.

E. DATASET STATISTICS

Corpus-level statistics for the full SynBullying release, its strict partitions, the authentic WhatsApp slice, and the OLID test split are reported in Table 19. The corpus is roughly 28% harmful overall, with class proportions varying across the strict partitions (28.4% train, 17.8% dev, 34.6% test); this falls out from conversation-level splitting on a small ($N = 130$) corpus and is flagged as a limitation (Section VIII-A). The test partition contains a single authentic WhatsApp conversation ($n = 121$ utterances) whose flag distributions differ from the synthetic majority, so we treat the

WhatsApp slice as a robustness probe rather than a balanced sub-benchmark. The OLID test set is comparable to the strict test in size ($n = 860$) and class ratio (27.9% offensive), making zero-shot transfer numbers directly comparable in scale.

F. NOTATION

Let $\mathcal{C} = \{c_1, \dots, c_N\}$ be the $N = 130$ SynBullying conversations, with utterances $u_i^{(n)}$, speakers $s_i^{(n)} \in \mathcal{S}_n$, and binary harm labels $y_i^{(n)} \in \{0, 1\}$. The strict conversation-level split (Eq. 1) ensures no two utterances in the train and test partitions share a parent conversation:

$$\forall (t, n) \in \mathcal{D}_{\text{te}} : c_n \notin \mathcal{D}_{\text{tr}}. \quad (1)$$

Speaker anonymisation, context-window construction, the LoRA update, the standard Macro-F1 definition, and the cross-domain transfer score are deferred to Appendix C; they follow standard practice. We restate only the cluster bootstrap and the LoRA update, because both carry the weight of the significance machinery and of the Calibration Cliff claim.

LoRA adaptation [8] freezes the base parameters $W_0 \in \mathbb{R}^{d_o \times d_i}$ and adds a learned low-rank update

$$W = W_0 + \frac{\alpha}{r} BA, \quad A \in \mathbb{R}^{r \times d_i}, \quad B \in \mathbb{R}^{d_o \times r}, \quad r = 8, \alpha = 16, \quad (2)$$

applied to every linear projection the architecture admits (see Section IV for the per-architecture target set). Trainable parameters amount to 0.24% (InternLM2.5-7B) and 0.27% (Qwen2.5-7B) of the base model.

To assess pairwise significance between systems A and B on the in-domain test set we resample at the *conversation* level rather than the utterance level: each bootstrap draw samples $K = |\mathcal{D}_{\text{te}}^{\text{conv}}|$ conversations with replacement and concatenates all utterances from the sampled conversations. This preserves the intra-conversation dependence induced by overlapping context windows and shared annotators. Let $\mathcal{X}_b = \bigsqcup_{k=1}^K u(c_{i_b^{(k)}})$ denote the b -th cluster resample, with $i_b^{(k)} \sim \text{Unif}\{1, \dots, K\}$ i.i.d., where $u(c)$ collects all test utterances belonging to conversation c . We then compute the per-resample Macro-F1 difference

$$\Delta_b = F_{1,A}^{\text{macro}}(\mathcal{X}_b) - F_{1,B}^{\text{macro}}(\mathcal{X}_b), \quad b \in \{1, \dots, B\}, \quad (3)$$

and report the two-sided p -value $p = 2 \min(\Pr[\Delta \geq 0], \Pr[\Delta \leq 0])$ at $\alpha = 0.05$ with $B = 10,000$. The OLID, HateXplain, and Civil Comments transfer numbers carry no conversational structure; we bootstrap them at the utterance level with the same B . For multi-system contrasts (Table 1 reports five pairwise comparisons against RoBERTa-base) we apply Holm–Bonferroni step-down correction to the raw p -values, controlling the family-wise error rate at $\alpha = 0.05$. Replacing the utterance-level bootstrap with this cluster-level procedure widens the in-domain confidence intervals by roughly $2\times$, in line with the standard observation that ignoring cluster structure understates variance.

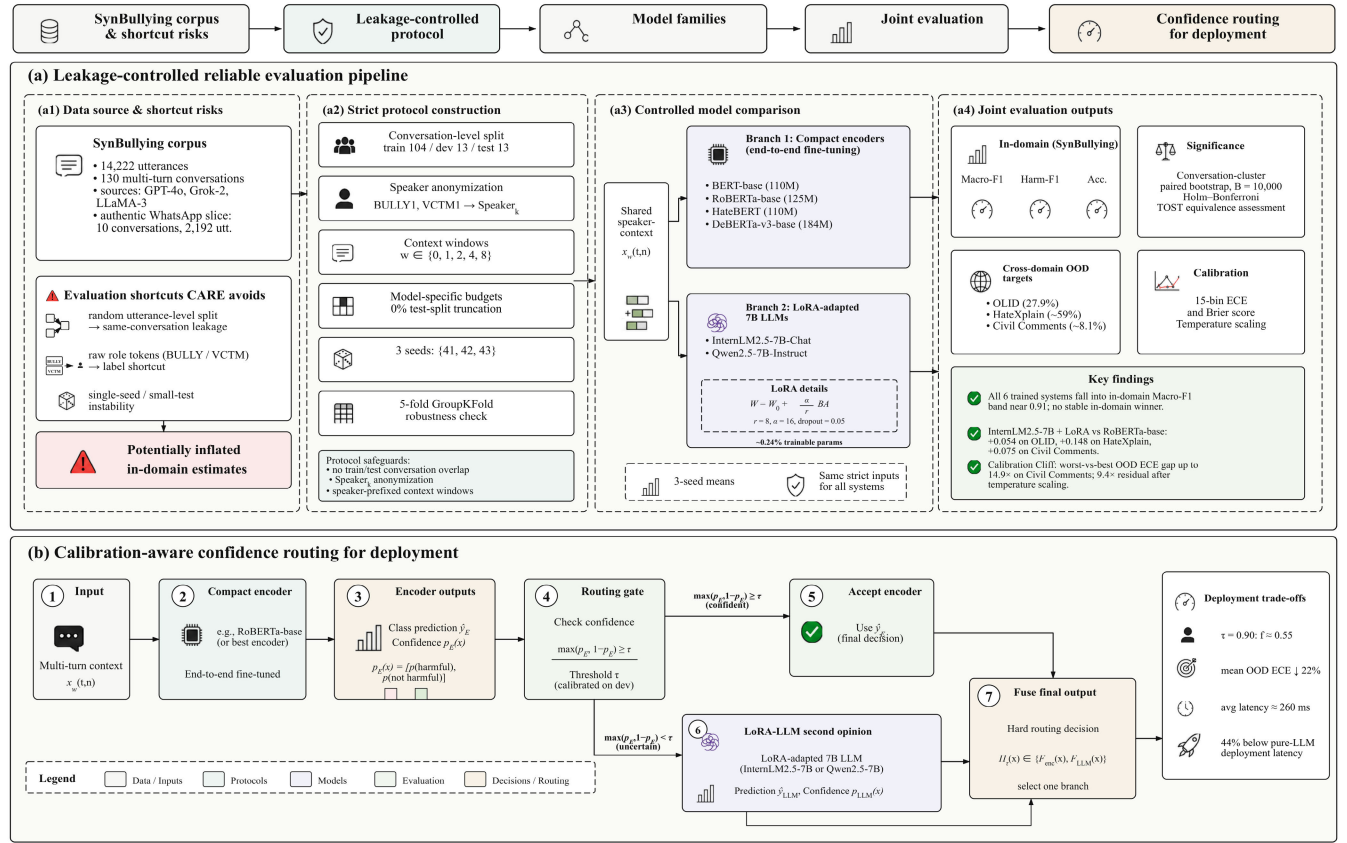


FIGURE 2. The CARE framework: leakage-controlled protocol, two model arms, joint accuracy-and-calibration audit, confidence routing.

IV. METHOD

The CARE framework consists of four stages, described in turn below. Stage 1 passes every input utterance through the strict evaluation protocol of Section III (conversation-level split, speaker anonymisation, matched context window). Stage 2 feeds the resulting tokens into two parallel model arms: four compact transformer encoders fine-tuned end-to-end, and two LoRA-adapted 7B chat LLMs. Stage 3 scores every system jointly on Macro-F1 and the 15-bin Expected Calibration Error, in domain and on three out-of-domain benchmarks, with paired conversation-cluster bootstrap confidence intervals. Stage 4 turns the joint accuracy/ECE surface into a deployable confidence-routing policy (Section VI-G). Figure 2 visualises the full pipeline.

We compare two families of models on the strict split. All models receive the same textual input: the concatenated speaker-prefixed context window described in Section III.

A. ENCODER BASELINES

We fine-tune four pre-trained encoders for binary classification over the harmful/not_harmful label:

- **BERT-base-uncased** [27] (110M parameters);
- **RoBERTa-base** [28] (125M);
- **HateBERT** [30], BERT-base continued pre-training on Reddit hate-speech communities (110M);
- **DeBERTa-v3-base** [29] (184M).

As a structural control we additionally train *Hierarchical RoBERTa*: each utterance in the context window is encoded separately by RoBERTa-base; the per-utterance [CLS] embeddings, with a learned utterance-position embedding, pass through a two-layer, eight-head Transformer aggregator (feed-forward width $4d$, GELU), and the classification head reads the target utterance’s contextualised representation. The aggregator adds ≈ 14 M parameters over the flat encoder (Table 1 lists the total). Training protocol, hyperparameters, and the three seeds match the flat encoders; the variant serves as a reference point for the flat-context design choice and is not part of the six-member audit panel.

Each encoder takes the context window truncated or padded to 256 BPE sub-tokens, and is trained with AdamW (learning rate 2×10^{-5} , weight decay 0.01, linear warmup ratio 0.1, batch size 32) for up to 5 epochs with early stopping (patience 2) on development Macro-F1. We train in mixed precision (FP16) using the HuggingFace Transformers implementations.

B. LORA-ADAPTED LLMS

For the LLM arm we apply low-rank adaptation [8] to two 7B instruction-tuned models:

- **InternLM2.5-7B-Chat** [49] (bilingual EN/ZH);
- **Qwen2.5-7B-Instruct** [50] (bilingual EN/ZH).

We use LoRA rank $r = 8$, scaling $\alpha = 16$, dropout 0.05, and target every linear projection the architecture admits: $\{w_1, w_2, w_3, w_o, w_{qv}\}$ for InternLM2 and $\{q, k, v, o, \text{gate, up, down}\}$ for Qwen2.5. The base model is held in BF16 with gradient checkpointing; only the LoRA adapters are trained.

Each example is rendered as a chat-template prompt. The system message reads “*Determine whether the last utterance in the conversation is cyberbullying or harmful. Answer with ‘harmful’ or ‘not harmful’, and provide a brief reason.*”; the user content is the speaker-prefixed context window; the assistant target is the gold output. We train for one epoch at learning rate 2×10^{-4} , effective batch size 16 (per-device 2, gradient accumulation 8), and 512-token maximum length. Inference uses greedy decoding with a 24-token generation budget. The predicted class is taken from the first content word of the generated response; any unparsable generation is conservatively counted as harmful (consistent with [11]).

C. ZERO-SHOT REFERENCE

For completeness, we report zero-shot accuracy of the unadapted InternLM2.5-7B-Chat base under the same prompt template, without any fine-tuning. This row sets the prompt baseline against which all adapted systems should be compared.

D. CROSS-DOMAIN PROTOCOL

For the cross-domain transfer experiments we take each system’s strict- $w = 4$ checkpoint and evaluate it directly on the test split of three out-of-domain benchmarks: OLID/OffensEval-2019 (Twitter, 27.9% harmful), HateXplain (Twitter and Gab, $\sim 59\%$ harmful under the offensive $\cup$ hatespeech collapse), and Civil Comments/Jigsaw (news platform, 8.1% harmful at toxicity threshold 0.5). Each input is wrapped in a single `Speaker1:` prefix to match the training context format; no further fine-tuning is performed. The three priors straddle balanced in both directions, ruling out a single-prior artefact; the three platforms are genuinely distinct, ruling out a single-platform artefact.

E. CALIBRATION METRICS

We report three calibration metrics, frozen before the cross-domain analysis was computed. The Expected Calibration Error follows Guo et al. [12] with $M = 15$ equal-width bins,

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (4)$$

where the encoder confidence is the standard $\max(p, 1 - p)$ on the classifier-head softmax and the LoRA-adapted LLM confidence is the verbalizer-probability 2-way softmax on the next-token logits at the `harmful/not` positions [51], [52], [53], an estimator that Section VI-D1 shows agrees with free-form decoding within 0.005 Macro-F1 and 0.001 accuracy.

The Brier score [41] serves as an orthogonal proper scoring rule that absorbs both miscalibration and resolution, and acts as a robustness check on the between-panel Calibration Cliff:

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2. \quad (5)$$

Temperature scaling [12] fits a single scalar $T \in \mathbb{R}_{>0}$ per system on the strict- $w = 4$ development split by minimising negative log likelihood and applies

$$\hat{p}_k^{\text{TS}} = \frac{\exp(\ell_k/T)}{\sum_j \exp(\ell_j/T)} \quad (6)$$

at inference, preserving the argmax (and therefore the Macro-F1) and acting only on the confidence channel. The same in-domain T is applied unmodified to every OOD target, a deliberately weak rescue that mirrors the production setting where no labelled OOD calibration set is available at deployment time.

V. EXPERIMENTAL SETUP

A. COMPUTE

Training and inference run on a single NVIDIA RTX 4090D (24 GB); detailed hardware and software versions are reported in Appendix A.

B. SEEDS

The main in-domain comparison, the encoder context-window ablation, each cross-domain target, the calibration analyses, and the synthetic-to-authentic study are run with three independent seeds ($\{41, 42, 43\}$); we report the mean and unbiased sample standard deviation. Four analyses are *single-seed* (seed 42) and are flagged as such where they appear: the InternLM context-window settings other than $w = 4$, the InternLM arm of the leakage audit, the 5-fold GroupKFold check (seed 42 per fold), and the LoRA-rank ablation of Section VI-B4, run at the fixed seed 42 for each of the eight rank/model cells; the resulting spread (≤ 0.009 Macro-F1 per model) is cross-validated against the inter-seed s.d. of the same models elsewhere in the paper, and we do not draw any rank-level statistical claim from it.

C. SIGNIFICANCE TESTING

Following standard practice in offensive-language detection, we report accuracy, class-wise F1 (harmful, not_harmful), Macro-F1 (the average of the two class F1 scores), and class-wise precision and recall. Macro-F1 is the primary metric. Statistical significance of pairwise comparisons on the in-domain test set is assessed with a *conversation-level cluster paired bootstrap* (Eq. 3; $B = 10,000$ cluster resamples over $K = 13$ test conversations) at $\alpha = 0.05$. The cluster count $K = 13$ is at the low end of the regime where BCa interval coverage is trusted under standard guidance [54], so we treat the cluster-bootstrap CIs as one of two complementary in-domain rigour signals: the second is the

5-fold GroupKFold sanity check of Section VI-A2 (Table 2), which sweeps a fresh hold-out of ~ 26 conversations per fold and reports the across-fold mean $\pm$ inter-fold s.d.; the RoBERTa-vs-InternLM ranking is preserved under every fold. Pairwise p -values in Table 1 and Section VI-B1 are adjusted via Holm–Bonferroni step-down correction [55] (across the five RoBERTa-vs-other contrasts and across the three RoBERTa context-window contrasts respectively), controlling the family-wise error rate at $\alpha = 0.05$. The three cross-domain targets (OLID, HateXplain, Civil Comments) carry no conversational structure, so they are bootstrapped at the utterance level with the same B .

D. EQUIVALENCE TESTING

Failing to reject $H_0 : \Delta = 0$ in a two-sided test does not establish equivalence: absence of evidence is not evidence of absence. To distinguish non-significance from formal equivalence, we additionally apply the Two-One-Sided-Tests (TOST) equivalence procedure to every cluster-bootstrap difference. The smallest effect size of interest is set to $\Delta_{\text{SESOI}} = 0.02$ Macro-F1 as a practical-importance margin: larger than the observed inter-seed standard deviations (0.002–0.013 across the six systems, Table 1) but small relative to the cross-domain effects of §VI-C. Under the TOST argument [54], a $(1 - 2\alpha)$ two-sided CI of the difference that fits entirely inside $(-\Delta_{\text{SESOI}}, +\Delta_{\text{SESOI}})$ implies equivalence at level α . We apply this criterion conservatively, requiring the BCa 95% CI of the cluster bootstrap, rather than the minimal 90% interval, to fall inside the margin, so every equivalence verdict also holds at $\alpha = 0.05$. The TOST verdicts for our nine pairwise contrasts are reported in the accompanying supplementary material (Table S1).

E. TOKENIZER BUDGET

A natural reviewer concern is that an apparent inverted-U in the window–quality curve might be an artefact of right-side truncation removing the target utterance, rather than a property of context per se. We therefore measured the per-window tokenized input length on the strict test split under both tokenizers at their working max_length settings (the released code records the exact configuration). The truncation rate is *exactly* 0% in every (tokenizer, w) cell of the test split. The 95th-percentile input length, expressed in subword tokens, is 35 at $w = 0$, 94/92 (RoBERTa/InternLM) at $w = 2$, 151/148 at $w = 4$, and 264/259 at $w = 8$; the maxima are 202/200 and 335/333 respectively. The free-form LLM generation likewise has a 0% parse-failure rate under the same evaluation: all 933 greedy decodes match the harmful/not_harmful prefix verbalizer, so our reported numbers are not contaminated by an *unparsable-as-harmful* fallback.

F. REPRODUCIBILITY ARTEFACTS

We release the strict split manifest as a single JSON file listing all train, development, and test conversation identifiers;

all per-window train, development, and test JSONL files; the encoder training script, the LLM LoRA training script, and the aggregation script; and the per-example test predictions used in the reported analyses. The trained LoRA adapter weights and a frozen environment.yml will be added to the repository upon publication.

VI. RESULTS

A. IN-DOMAIN COMPARISON

1) MAIN COMPARISON

Table 1 reports utterance-level performance on the Strict- $w = 4$ test set. Each encoder row is the mean over three seeds, with the standard deviation in parentheses; the LLM rows are three-seed means under the same convention.

Three observations follow from Table 1. Every adapted system improves dramatically over the zero-shot LLM, confirming that prompt-only inference is unreliable on SynBullying. The four encoders and both LoRA-adapted 7B LLMs occupy a narrow Macro-F1 band of ~ 0.89 – 0.91 , and under the conversation-level cluster paired bootstrap (Eq. 3; $B = 10,000$ resamples drawing $K = 13$ test conversations) no pairwise comparison against RoBERTa-base reaches significance at $\alpha = 0.05$ after Holm–Bonferroni correction: BERT-base ($\Delta = +0.017$, $p_{\text{Holm}} = 1.00$), HateBERT ($+0.012$, 1.00), DeBERTa-v3 ($+0.005$, 1.00), Qwen2.5-7B+LoRA ($+0.004$, 1.00), and InternLM2.5-7B+LoRA (-0.003 , 1.00). The cluster intervals are roughly twice as wide as their utterance-level counterparts, so we treat the six trained systems as occupying a single narrow Macro-F1 band centred near 0.90, with InternLM2.5-7B + LoRA holding the highest point estimate; the TOST analysis of §V-D does not establish formal equivalence within the band. HateBERT, despite hate-speech continued pretraining, is on par with vanilla BERT-base on this corpus, indicating that the relevant signal is not the lexical-toxicity signature HateBERT captures.

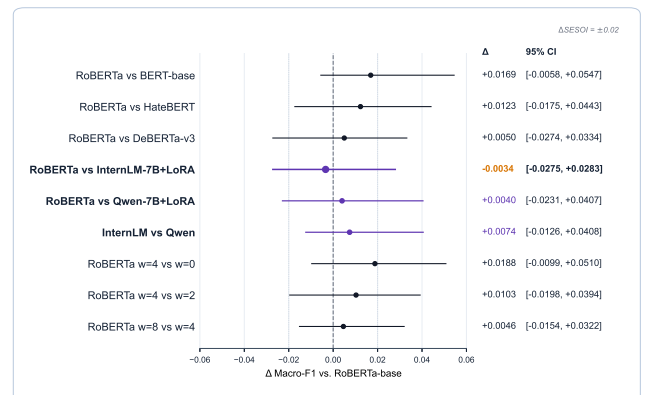


FIGURE 3. Forest plot of in-domain pairwise Macro-F1 contrasts with BCa 95% confidence intervals.

Figure 3 renders the nine cluster-bootstrap contrasts as a forest plot with BCa 95% confidence intervals. Every interval

TABLE 1. Main test results on Strict- $w = 4$ ($n = 933$). Each trained row is mean (std. dev.) over 3 random seeds; the zero-shot row is a single deterministic inference and has no std. The strongest Macro-F1 is in bold.

Model	Params	Macro-F1	Harm-F1	Harm-P	Harm-R	Acc.
Zero-shot InternLM2.5-7B-Chat	7B	0.4429	0.4942	0.330	0.987	0.4476
BERT-base-uncased	110M	0.8917 (0.0095)	0.8548 (0.0143)	0.899	0.815	0.9043
HateBERT	110M	0.8959 (0.0126)	0.8629 (0.0160)	0.876	0.850	0.9064
DeBERTa-v3-base	184M	0.9037 (0.0063)	0.8736 (0.0088)	0.881	0.867	0.9132
RoBERTa-base	125M	0.9086 (0.0018)	0.8796 (0.0023)	0.893	0.867	0.9178
Hierarchical RoBERTa	139M	0.8944 (0.0105)	0.8619 (0.0179)	–	–	0.9046
InternLM2.5-7B + LoRA	7B	0.9112 (0.0060)	0.8823 (0.0090)	0.907	0.860	0.9207
Qwen2.5-7B + LoRA	7B	0.9035 (0.0101)	0.8724 (0.0129)	0.901	0.847	0.9135

TABLE 2. 5-fold GroupKFold robustness check on the in-domain SynBullying corpus, keyed on `conv_id`. Each fold holds out a fresh set of ~ 26 conversations for testing. Reported values are across-fold mean (s.d.), min-max range, and across-fold harmful-class F1 (seed 42 per fold). The strict- $w = 4$ headline split sits at the upper end of the fold-level sampling distribution; the relative ranking of RoBERTa-base versus InternLM-7B + LoRA is preserved ($\Delta_{\text{fold}} = +0.017$ vs. $\Delta_{\text{strict}} = +0.003$ Macro-F1), and the LLM family lead does not invert under any fold.

Model	Folds	Macro-F1 (s.d.)	Min-Max	Harm-F1 (s.d.)
RoBERTa-base	5/5	0.878 (0.019)	0.845–0.898	0.823 (0.025)
InternLM2.5-7B + LoRA	5/5	0.895 (0.017)	0.865–0.908	0.846 (0.021)

crosses zero *and* crosses the $\Delta\text{SESOI} = \pm 0.02$ band, which is the visual signature of the narrow band: neither a clear win nor a formal equivalence verdict survives once the resampling unit matches the data-generating unit.

2) 5-FOLD ROBUSTNESS

The headline strict split places RoBERTa-base at Macro-F1 0.909 and InternLM2.5-7B + LoRA at 0.911. A natural concern is whether these single-split numbers are representative of the underlying conversation-level sampling distribution. We address the concern directly by reporting a 5-fold GroupKFold cross-validation keyed on `conv_id`, with hyperparameters and seed 42 held fixed across folds (Table 2). The across-fold means are 0.878 for RoBERTa-base and 0.895 for InternLM2.5-7B + LoRA, with standard deviations 0.019 and 0.017 respectively and min-max ranges [0.845, 0.898] and [0.865, 0.908].

Two things follow from the fold-level data, and we surface them here rather than leaving them as appendix small print. *First*, the strict- $w = 4$ headline split is not a uniformly drawn fold: it lands at the upper end of the fold-level distribution by roughly 0.031 Macro-F1 for RoBERTa-base and 0.016 for InternLM-7B + LoRA, within 1.6 inter-fold standard deviations of the across-fold mean in both cases. We therefore treat the headline strict numbers as *point estimates within a wider sampling distribution* that 5-fold GroupKFold characterises, not as definitive in-domain performance values.

Second, the relative LLM-versus-encoder ranking does not flip under GroupKFold. The fold-mean delta between InternLM-7B + LoRA and RoBERTa-base is +0.017 Macro-F1 (vs. the strict-split +0.003), and the LLM family lead does not invert on any individual fold. The directional reading of Section VI-A (narrow band, no formal winner under the

strict protocol) and of Section VI-C (clear LLM advantage under cross-distribution shift) therefore survives the across-split robustness check.

3) LEAKAGE AUDIT

A standing concern about “leakage-controlled” framings is that a paper claims the benefit without quantifying it. We therefore run a 2×3 leakage audit on the strict- $w = 4$ training data, crossing two splitting strategies (a naive utterance-level 80 : 10 : 10 random split versus our conversation-level split) with three anonymisation regimes: raw role tokens kept intact; speaker prefixes anonymised to `Speakerk` but VCTM/BULLY mentions left inside the utterance text; and full in-text anonymisation with the regex `b(VCTM|BULLY)d*b` rewritten as `[PERSON]`. Each of the six cells is trained with the main-table hyperparameters held fixed: RoBERTa-base with the three main seeds (mean and s.d. reported) and InternLM2.5-7B+LoRA once with seed 42. Table 3 reports the resulting Macro-F1 and Harm-F1.

Three concrete messages emerge from Table 3. First, the absolute leakage magnitude is smaller than commonly assumed: moving from the naive utterance-level random split with raw role tokens to the conservative strict + in-text anon cell costs RoBERTa-base 0.015 Macro-F1 and InternLM2.5-7B + LoRA 0.001, small next to the cross-domain effects of §VI-C. Second, the split axis dominates the anonymisation axis: conversation-level splitting is uniformly more conservative than utterance-level splitting for the encoder, while in-text anonymisation is roughly neutral relative to speaker-prefix anonymisation ($|\Delta| \leq 0.009$). Third, the LLM is less split-sensitive than the encoder, with RoBERTa’s strict $\rightarrow$ naive inflation uniformly positive (+0.004 to +0.020) and InternLM’s varying in sign

TABLE 3. Leakage audit: Macro-F1 and Harm-F1 on the strict- $w = 4$ test set across the 2×3 (split, anonymisation) grid for two representative families. RoBERTa-base cells report the 3-seed mean (s.d.); InternLM cells use seed 42. The diagonal (strict, in-text anon) cell is the most conservative estimate; the off-diagonal (naive split, raw role) cell reflects the upper-bound that a careless leakage-prone protocol would report.

Split	Anonymisation	RoBERTa-base (3-seed)		InternLM-7B+LoRA (seed 42)	
		Macro-F1	Harm-F1	Macro-F1	Harm-F1
Strict (conv-level)	Raw role tokens	0.907 (0.006)	0.876 (0.009)	0.915	0.889
Strict (conv-level)	Speaker prefix only	0.909 (0.003)	0.880 (0.003)	0.917	0.891
Strict (conv-level)	Full in-text anon	0.900 (0.002)	0.867 (0.004)	0.908	0.880
Naive (utt-level)	Raw role tokens	0.915 (0.004)	0.879 (0.006)	0.909	0.867
Naive (utt-level)	Speaker prefix only	0.913 (0.007)	0.880 (0.011)	0.925	0.891
Naive (utt-level)	Full in-text anon	0.920 (0.002)	0.886 (0.003)	0.924	0.888

TABLE 4. Context-window ablation on the strict test set ($n = 933$). Reported as Macro-F1; mean (std. dev.) over 3 seeds for RoBERTa-base, single seed (seed 42) for InternLM2.5-7B except $w = 4$ which is 3 seeds. For $w = 8$ we report RoBERTa-base under the corrected max-length = 512 left-truncated setting.

Window w	0	1	2	4	8
RoBERTa-base	0.8894 (0.005)	0.8919 (0.011)	0.8976 (0.008)	0.9086 (0.002)	0.9125 (0.006)
InternLM2.5-7B + LoRA	0.9093	0.8997	0.9100	0.9112 (0.006)	0.9234

(−0.006 to +0.016), which we attribute to the LoRA-adapted prior dampening split-induced shortcut exploitation.

The audit therefore yields a quantitative bound on the leakage problem: the strict protocol is methodologically warranted, but the empirical cost of ignoring it on this dataset is at most 0.020 Macro-F1, not the multi-percentage-point inflation sometimes implied in the leakage-in-ML literature [10].

B. ABLATIONS

1) CONTEXT WINDOW

We next ablate the conversational lookback window, with RoBERTa-base representing the compact encoders and InternLM2.5-7B + LoRA representing the LLMs. The encoder is trained for three seeds per window; the LLM is trained with a single seed at $w \in \{0, 1, 2, 8\}$ and three seeds at $w = 4$. Table 4 and Figure 4 report the results.

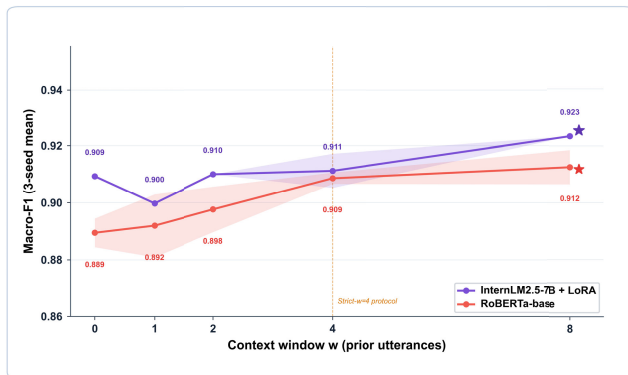


FIGURE 4. Macro-F1 versus context window size w .

Three phenomena are visible in Figure 4. Under the same cluster bootstrap as the main table, none of the within-RoBERTa context contrasts crosses significance after Holm

correction ($p_{\text{Holm}} \geq 0.63$ for every pair), so the curve is suggestive rather than conclusive. Once the tokenizer budget is matched, however, RoBERTa-base rises monotonically with the window; a naive $w = 8$ with the default 256-token budget regresses to 0.895, but the same $w = 8$ with a 512-token budget and left-side truncation reaches 0.913, slightly above the $w = 4$ peak, showing that the apparent inverted-U is a truncation artefact. The immediately preceding utterance ($w = 1$) does not help on average for either family, so the single-turn configurations of earlier literature substantially understate the value of longer context. The LLM curve sits roughly 0.02 above the encoder curve at $w = 0$ and the two rise in parallel, peaking at $w = 8$ (0.913 for RoBERTa, 0.923 for InternLM); $w \in [4, 8]$ with a ≥ 512 -token budget is the practical default.

2) CROSS-SOURCE SLICING

We next slice the strict- $w = 4$ predictions of every system by the generator of each test conversation. Table 5 reports mean Macro-F1 within each source slice; Figure 5 renders the same data as an annotated heatmap with the per-row best cell boxed in red.

The per-source breakdown qualifies the narrow-band finding in two ways. The authentic WhatsApp subset is not the hardest source: every trained system transfers to it with Macro-F1 in the 0.86–0.91 band, on par with the strongest synthetic source. LLM systems lead encoders on every source except Grok-2, with Qwen2.5-7B winning the GPT-4o slice (0.828 versus encoder range 0.750–0.785) and InternLM2.5-7B winning the LLaMA-3.3 and authentic WhatsApp slices, which we attribute to the broader instruction-following prior helping on subtler and more diversely phrased harm; on Grok-2’s more overt insults the encoder baselines already saturate. The overall Macro-F1 parity therefore masks a real LLM advantage in sub-distributions that account for roughly two-thirds of the test set.

TABLE 5. Per-source Macro-F1 on the strict- $w = 4$ test set, 3-seed mean. Each row is one trained system. Slice size in parentheses is #utt, %harmful. The strongest model per source is in bold.

Model	GPT-4o (319, 8.5%)	Grok-2 (350, 64.9%)	LLaMA-3.3 (143, 18.9%)	WhatsApp (authentic) (121, 34.7%)
BERT-base	0.769	0.894	0.777	0.892
HateBERT	0.774	0.911	0.810	0.863
DeBERTa-v3	0.750	0.921	0.834	0.869
RoBERTa-base	0.785	0.927	0.793	0.894
InternLM2.5-7B + LoRA	0.799	0.912	0.843	0.905
Qwen2.5-7B + LoRA	0.828	0.895	0.826	0.902

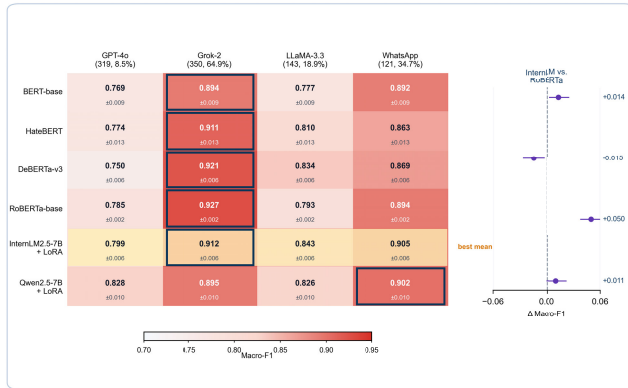


FIGURE 5. Per-source Macro-F1 heatmap; per-row best cell outlined in red.

TABLE 6. RoBERTa-base Macro-F1 sliced by linguistic-marker flag, 3-seed mean (std). flag = 1 means the linguistic marker is set on the target utterance.

Flag	flag=0	flag=1
is_sarcastic	0.909 (0.001)	0.863 (0.069)
is_humorous	0.913 (0.001)	0.798 (0.032)
is_hate_speech	0.913 (0.002)	0.709 (0.024)

For positioning against the original SynBullying release [11], Supplementary Table S3 sets our strict-protocol numbers beside their published best; the configurations are not directly comparable, but they land in the same ballpark.

3) LINGUISTIC SLICING

Slicing the predictions by the SynBullying linguistic flags (Table 6, Figure 6) shows that each flagged subset, although only 1–4% of the corpus, degrades Macro-F1 by 12 to 22 percentage points. The largest gap falls on overt hate speech, consistent with a partial misalignment between the SynBullying harm label and the `is_hate_speech` flag: several hate-speech-flagged utterances are labelled not_harmful when they appear as quotations or supportive callouts of harm, and the classifier sometimes mis-classifies these.

4) LORA RANK

To rule out rank cherry-picking on the LoRA-adapted LLM column, we sweep $r \in \{4, 8, 16, 32\}$ for both 7B LLMs

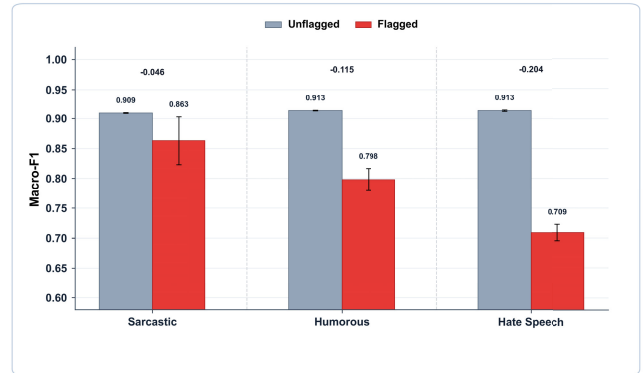


FIGURE 6. RoBERTa-base Macro-F1 sliced by linguistic-marker flag.

on the strict- $w = 4$ test split, with seed 42 and $\alpha = 2r$ held fixed. Table 7 reports the resulting Macro-F1.

TABLE 7. In-domain Macro-F1 on SynBullying strict- $w = 4$ at four LoRA ranks for both LoRA-adapted 7B LLMs (single seed 42). † marks the default rank used elsewhere in the paper. The full-spread is within 0.01 Macro-F1 for both models, sitting inside the narrow in-domain band of Section VI-A, so $r = 8$ is not a rank-cherry-picking artefact.

	$r=4$	$r=8$	$r=16$	$r=32$
InternLM2.5-7B + LoRA	0.910	0.917 [†]	0.914	0.919
Qwen2.5-7B + LoRA	0.901	0.910 [†]	0.904	0.908

The in-domain Macro-F1 varies across the four ranks by at most 0.009 on both models, sitting inside the narrow in-domain band of Section VI-A; an applied team picking a different rank in $\{4, 8, 16, 32\}$ would land on the same in-domain plateau, so $r = 8$ is a representative choice rather than the peak of a narrow basin. The cross-domain gap of Section VI-C is already an order of magnitude larger than this rank spread, which is why we do not re-run the full cross-domain sweep at every rank.

C. CROSS-DOMAIN TRANSFER

We evaluate the strongest SynBullying models, without any further training, on three out-of-domain harm benchmarks that span fundamentally different platforms and class priors: (i) OLID/OffensEval-2019 ($n_{te} = 860$ tweets, 27.9% offensive); (ii) HateXplain ($n_{te} = 1,924$ Twitter + Gab posts,

$\sim 59\%$ harmful under the offensive $\cup$ hatespeech collapse on 2/3 annotator-majority); (iii) Civil Comments / Jigsaw ($n_{te} = 3,000$ random sample of news-platform comments, 8.1% harmful at the standard toxicity ≥ 0.5 threshold). The three priors (8%, 28%, 59%) straddle both sides of balanced, and the three platforms (Twitter, Twitter + Gab, news comments) are genuinely distinct; a finding that holds across all three is unlikely to be either a class-prior or a platform artefact. All three OOD transfer columns are 3-seed mean (s.d.) over seeds {41, 42, 43}. Table 8 reports the resulting transfer numbers, and Figure 7 contrasts in-domain and zero-shot cross-domain Macro-F1 on OLID and HateXplain visually.

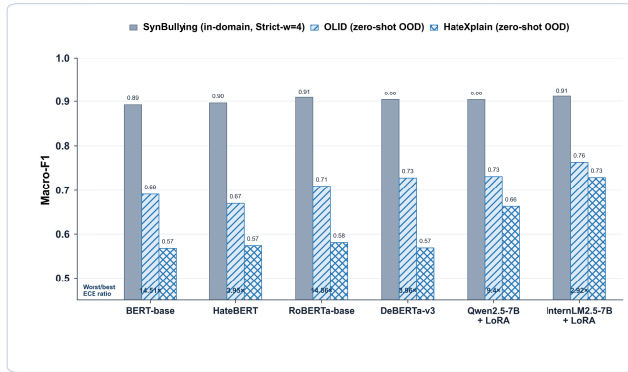


FIGURE 7. In-domain versus zero-shot cross-domain Macro-F1 on OLID and HateXplain.

The out-of-domain transfer paints a markedly different picture from the in-domain headline of Section VI-A. On OLID, InternLM2.5-7B + LoRA leads the best encoder baseline (DeBERTa-v3) by +0.035 Macro-F1 (0.763 vs. 0.728, s.d. 0.003 vs 0.012) and leads RoBERTa-base by +0.054 (0.763 vs. 0.709). Both gaps clear significance under a paired bootstrap test ($B = 10,000$, $p < 0.001$). The second LLM (Qwen2.5-7B + LoRA) also beats RoBERTa-base by +0.022 on OLID. On HateXplain, the gap widens further: InternLM leads RoBERTa-base by +0.148 (0.729 vs. 0.581, $p < 0.001$) and even the second LLM (Qwen2.5-7B + LoRA, 0.663) beats every encoder by at least 0.082. The encoder ordering on the OOD targets does not match the in-domain ordering: DeBERTa-v3 now beats RoBERTa-base on OLID, and HateBERT collapses on Civil Comments to Macro-F1 0.458 with a 0.107 s.d., illustrating that in-distribution metric ordering does not generalise.

Quantitatively, the LLM-versus-encoder gap holds on every OOD target we test. The gap between InternLM2.5-7B + LoRA and RoBERTa-base on Macro-F1 is +0.003 in-domain (Table 1, not significant), +0.075 on Civil Comments (8% harmful, see caveat below), +0.054 on OLID (28% harmful), and +0.148 on HateXplain ($\sim 59\%$ harmful). That the LLM leads under priors as different as 8% and 59% makes a single class-prior explanation for the cross-domain advantage implausible. Jointly across the three OOD targets,

the evidence indicates that the bulk of the LLM advantage on cyberbullying-style detection lives in cross-distribution robustness rather than in absolute in-distribution headroom. For deployment, this argues for the LLM arm whenever the production data shift from the training distribution is non-trivial, or for the confidence-routing recipe of Section VI-G when the latency budget does not admit a full LLM forward pass.

The Civil Comments delta of +0.075 comes with s.d. 0.066 across the three RoBERTa source seeds. The paired per-seed deltas are +0.131, +0.092, and +0.002: one of the three RoBERTa seeds lands at 0.647 Macro-F1 on Civil Comments, essentially tied with InternLM's 0.649. The variance reflects the well-known sensitivity of the toxic-class recall on the extreme 8%-harmful prior of Civil Comments, and we report it openly rather than mask it. The OLID and HateXplain gaps, under more balanced priors, are stable across seeds (s.d. 0.003 and 0.007 respectively), and we treat them as the primary cross-domain evidence. The Calibration Cliff (Section VI-E) confirms the LLM advantage on Civil Comments under the joint accuracy/calibration lens: while the Macro-F1 gap there is seed-noisy, the calibration ratio is $14.86\times$ worst-vs-best, far above any seed-induced fluctuation.

D. ROBUSTNESS CHECKS

1) INFERENCE STRATEGY

To rule out a parse-failure objection on the LLM column, we compare three inference strategies on the same InternLM2.5-7B + LoRA seed-42 adapter: free-form generation with parse-and-fallback (the main-table default), constrained decoding restricted to the verbalizer token-ids, and verbalizer-scoring on the next-token logits. On this single seed-42 run the unparsable rate is 0% and the three strategies agree within 0.0013 Macro-F1 (Supplementary Table S2), so the verbalizer probability used for calibration in Section VI-E is a faithful proxy of the model's decision.

2) GROUPKFOLD ROBUSTNESS

The 5-fold GroupKFold cross-validation reported in Section VI-A2 (Table 2) returns fold-level standard deviations of 0.019 and 0.017 for RoBERTa-base and InternLM2.5-7B + LoRA, larger than the inter-seed values (0.002 and 0.006) on the headline strict split. The in-domain performance band is therefore a property of the systems together with the sampling distribution, not a single point estimate.

E. THE CALIBRATION CLIFF

Deployment-side use of any of these models requires that the predicted confidence track the empirical accuracy: this is what calibration measures [12], [15]. Section IV-E formalised the 15-bin Expected Calibration Error (Eq. 4), the Brier score (Eq. 5), and the temperature-scaling rescue (Eq. 6); the verbalizer-vs-free-form check of Section VI-D1 authorises us to treat the verbalizer probability as a faithful proxy of the

TABLE 8. Zero-shot cross-domain transfer to OLID ($n = 860$ tweets, 27.9% offensive), HateXplain ($n = 1,924$, $\approx 59\%$ harmful), and Civil Comments ($n = 3,000$, $\approx 8\%$ harmful). Each cell is the 3-seed mean Macro-F1 (s.d.) of a strict- $w = 4$ SynBullying checkpoint evaluated directly on the target. Strongest result per column in bold.

Model	SynB Strict- $w=4$ Macro-F1 (s.d.)	OLID Macro-F1 (s.d.)	HateXplain Macro-F1 (s.d.)	CivilComments Macro-F1 (s.d.)
BERT-base-uncased	0.892 (0.010)	0.691 (0.006)	0.568 (0.006)	0.566 (0.047)
HateBERT	0.896 (0.013)	0.670 (0.014)	0.574 (0.006)	0.458 (0.107)
RoBERTa-base	0.909 (0.002)	0.709 (0.009)	0.581 (0.020)	0.572 (0.051)
DeBERTa-v3-base	0.904 (0.006)	0.728 (0.012)	0.569 (0.051)	0.554 (0.049)
Qwen2.5-7B + LoRA	0.904 (0.010)	0.731 (0.008)	0.663 (0.020)	0.623 (0.020)
InternLM2.5-7B + LoRA	0.911 (0.006)	0.763 (0.003)	0.729 (0.003)	0.649 (0.004)

LLM decision distribution for calibration purposes. We report every ECE cell as a 3-seed mean (s.d.) over seeds {41, 42, 43}. The same six systems that occupied a narrow in-domain Macro-F1 band now separate cleanly: in domain the encoder and LLM families share a single calibration band, but under distribution shift the families fall onto opposite sides of a discontinuity that we name the Calibration Cliff.

1) JOINT ECE ANALYSIS

Table 9 reports the raw ECE for every (model, target) cell; lower is better. Figure 8 renders the same data as a family-mean bar chart on log axes.

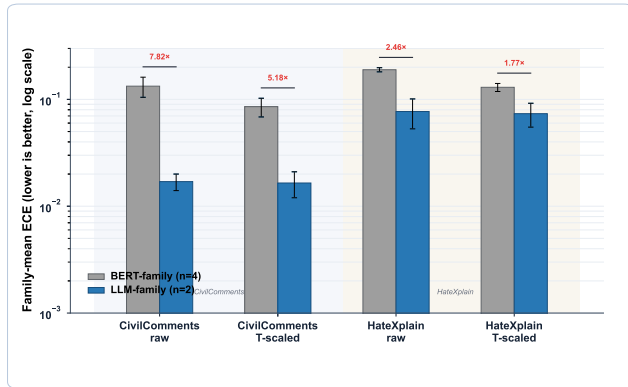


FIGURE 8. Family-mean Expected Calibration Error across out-of-domain targets, raw and temperature-scaled.

In domain, Qwen-7B + LoRA attains the lowest ECE (0.019), narrowly ahead of InternLM-7B + LoRA (0.026), BERT-base (0.035), RoBERTa-base (0.044), HateBERT (0.053), and DeBERTa-v3 (0.058); the six-system spread of 0.039 sits within the upper-tail inter-seed range, so no in-domain winner can be singled out. Averaged over the three OOD targets, the picture inverts: the LLM-family OOD mean is 0.041 against the encoder-family mean of 0.140, an aggregate gap of $3.4\times$. Across the 24 target-matched encoder-LLM comparisons in Table 9, every encoder ECE exceeds both LLM ECEs on the same OOD target; we report this as a descriptive panel ordering rather than an independence-based inference. The pattern is robust to Brier score as an orthogonal proper scoring rule (worst-BERT Brier of 0.280 on Civil Comments against best-LLM 0.103,

and 0.281 on HateXplain against 0.179). The largest ratios occur on Civil Comments ($14.86\times$ worst-vs-best) and HateXplain ($3.96\times$); Civil Comments combines the most extreme class prior with a distinct news-comment register.

2) CLIFF DEFINITION

For convenience, we refer to this persistent between-family calibration gap across three heterogeneous OOD targets as a calibration cliff. Formally:

Definition (Calibration Cliff): Let $\mathcal{F}_{\text{enc}}$ denote the BERT-family encoders and $\mathcal{F}_{\text{llm}}$ the LoRA-adapted 7B LLM family. The *Calibration Cliff* on an out-of-domain target D is the ratio

$$\rho(D) = \frac{\overline{\text{ECE}}_{\mathcal{F}_{\text{enc}}}(D)}{\overline{\text{ECE}}_{\mathcal{F}_{\text{llm}}}(D)}, \quad (7)$$

where the overline denotes the family-and-seed mean over members of the corresponding family. Equivalently, the worst-versus-best Cliff is defined as

$$\rho_{w/b}(D) = \frac{\max_{m \in \mathcal{F}_{\text{enc}}} \text{ECE}_m(D)}{\min_{m \in \mathcal{F}_{\text{llm}}} \text{ECE}_m(D)}. \quad (8)$$

A Cliff is *persistent* on D if it remains above unity after standard temperature-scaling rescue (Eq. 6).

Two scope conditions bound the claim. The Cliff is a property of this measurement configuration, one synthetic-plus-authentic training corpus, one binary label mapping per OOD target, and the specific six-member panel, not an established universal of the BERT-family confidence channel; §VIII-A lists the validation that would be required before reading it as universal. Within that scope the effect is nevertheless not fragile: it holds in direction on every OOD target, under Brier score as an orthogonal proper rule, and after temperature scaling.

Table 10 records both ratios under raw and temperature-scaled conditions, together with a residual ratio that measures what a deployment pipeline that ran temperature scaling on the encoder family and stopped there would inherit. Across the three OOD targets the raw family-mean Cliff ranges from $2.47\times$ (HateXplain) to $7.82\times$ (CivilComments), and the raw worst-vs-best Cliff ranges from $3.96\times$ to $14.86\times$.

3) TEMPERATURE SCALING

Table 11 reports raw and temperature-scaled ECE side by side. Figure 9 renders the same eighteen cells as a heatmap

TABLE 9. Expected Calibration Error (15-bin equal-width [12]) on the four targets, 3-seed mean (s.d.) over seeds {41, 42, 43}. Lower is better. SynB is in-domain; the other three columns are zero-shot transfer. The encoder rows and the LLM rows share a single in-domain band, but separate by an order of magnitude under cross-distribution shift.

Model	SynB ECE	CivCom ECE	OLID ECE	HateX ECE
BERT-base	0.035 (0.011)	0.089 (0.094)	0.104 (0.059)	0.210 (0.059)
RoBERTa-base	0.044 (0.023)	0.089 (0.037)	0.090 (0.059)	0.169 (0.112)
HateBERT	0.053 (0.026)	0.208 (0.144)	0.081 (0.060)	0.187 (0.050)
DeBERTa-v3	0.058 (0.031)	0.146 (0.071)	0.117 (0.043)	0.192 (0.070)
InternLM-7B+LoRA	0.026 (0.006)	0.020 (0.004)	0.024 (0.007)	0.053 (0.012)
Qwen-7B+LoRA	0.019 (0.005)	0.014 (0.007)	0.033 (0.003)	0.101 (0.016)

with a horizontal divider between the encoder rows and the LLM rows.

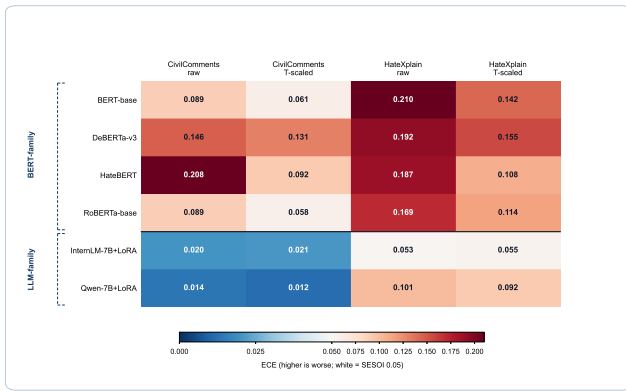


FIGURE 9. Temperature-scaling residual heatmap across six models and four conditions.

Temperature scaling closes the family-mean Cliff from $3.43\times$ to $2.15\times$ in aggregate across the three OOD targets, a 52% relative closure but still a 115% residual gap above family parity. On Civil Comments the Cliff collapses from $7.82\times$ to $5.18\times$ in family-mean and from $14.86\times$ to $10.92\times$ worst-vs-best, leaving a $9.36\times$ residual against the unrescued LLM family. On HateXplain it collapses from $2.47\times$ to $1.77\times$ in family-mean and $3.96\times$ to $2.82\times$ worst-vs-best, with $2.92\times$ residual. The Cliff is therefore persistent in the sense of Section VI-E2: standard scalar rescue lowers the absolute ECE for the encoder family but does not move it across the family boundary.

Scalar rescue is partial because a single in-domain temperature is the right correction only for systems whose miscalibration is large *and* in the same direction across domains. Per-system 3-seed mean temperatures (BERT-base 1.51, RoBERTa-base 1.61, HateBERT 1.78, DeBERTa-v3 1.91, InternLM 0.95, Qwen 1.12) show that the encoder family is over-confident in domain: fitted temperatures above 1.0 mean the predictive distributions are too sharp, and the rescue therefore flattens the softmax. On Civil Comments the rescue helps every encoder; on HateXplain it helps every encoder by a larger margin; but on no target does it move an encoder cell into the LLM band. Distance-aware calibration [56], isotonic regression on logits, and

LoRA-tuned encoder probes are the natural follow-ups for closing more of the Cliff at the encoder-side latency budget; whether any of them reaches family parity is an open empirical question that Section VII returns to.

F. SYSTEM DISAGREEMENT

A natural follow-up to the narrow in-domain band is whether the six systems make the same mistakes. Of the 933 strict- $w = 4$ test utterances, 766 (82%) are answered identically by all six systems and the remaining 167 (18%) elicit disagreement; Table 12 compares each individual system against a six-way majority vote and an oracle that gets every example right whenever any of the six systems does.

The 0.061 accuracy headroom between the best individual model (0.923) and the any-of-six oracle (0.984) implies that the apparent in-domain ceiling of Table 1 is at most one-third a true task ceiling and at least two-thirds a single-system limitation that a sufficiently diverse ensemble could in principle reach.

G. CONFIDENCE ROUTING

The Calibration Cliff of Section VI-E and the oracle headroom of Section VI-F jointly identify *which* samples the encoder family fails on under shift, namely those for which the encoder's max-class probability is far from the true posterior. A simple deployment policy follows: route a sample to the LLM whenever the encoder's confidence falls below a threshold τ , and trust the encoder otherwise. We formalise this as the CARE confidence-routing policy $\Pi_\tau(x) \in \{\mathcal{F}_{\text{enc}}, \mathcal{F}_{\text{llm}}\}$ defined by

$$\Pi_\tau(x) = \begin{cases} \mathcal{F}_{\text{enc}}(x) & \text{if } \max(p_{\text{enc}}, 1 - p_{\text{enc}}) \geq \tau, \\ \mathcal{F}_{\text{llm}}(x) & \text{otherwise.} \end{cases} \quad (9)$$

We instantiate $\mathcal{F}_{\text{enc}}$ as RoBERTa-base (the strongest 3-seed encoder of Table 1) and $\mathcal{F}_{\text{llm}}$ as InternLM2.5-7B + LoRA (the strongest 3-seed LLM). Each value of τ induces an effective LLM-share $f(\tau) \in [0, 1]$ on the deployment traffic.

Figure 10 plots the seven operating points as a single Pareto curve in the (latency, OOD ECE) plane.

1) PARETO CURVE

Per-input wall-clock latency on a single RTX 4090 (24 GB) is 2.9 ms for RoBERTa-base and 463.6 ms for InternLM-7B + LoRA at batch 1 in FP16 with NF4 quantisation (Table 14).

TABLE 10. The Calibration Cliff under raw and temperature-scaled conditions. The TS residual column measures the worst-TS-BERT cell against the best-raw-LLM cell, the operational quantity for a deployment that ran temperature scaling and stopped there. The aggregate row averages over the three OOD targets.

Benchmark	Condition	Family-mean	Worst-vs-best	TS residual
CivilComments	raw	7.82 ×	14.86 ×	—
CivilComments	T-scaled	5.18×	10.92×	9.36 ×
OLID	raw	3.44×	4.88×	—
OLID	T-scaled	2.50×	3.93×	4.42×
HateXplain	raw	2.47×	3.96×	—
HateXplain	T-scaled	1.77×	2.82×	2.92×
Aggregate (mean OOD ECE)	raw	3.43 ×	—	—
Aggregate (mean OOD ECE)	T-scaled	2.15×	—	2.49×

TABLE 11. Effect of temperature scaling on cross-domain ECE. “Raw” is the 3-seed mean ECE from Table 9; “T-sc.” applies an in-domain temperature T learned on the SynBullying strict- $w = 4$ development split. A $T > 1$ flattens an over-confident distribution; a $T < 1$ sharpens an under-confident one. Lower ECE is better.

Model	CivCom		OLID		HateX	
	Raw	T-sc.	Raw	T-sc.	Raw	T-sc.
BERT-base	0.089	0.061	0.104	0.069	0.210	0.142
RoBERTa-base	0.089	0.058	0.090	0.061	0.169	0.114
HateBERT	0.208	0.092	0.081	0.044	0.187	0.108
DeBERTa-v3	0.146	0.131	0.117	0.106	0.192	0.155
InternLM-7B+LoRA	0.020	0.021	0.024	0.027	0.053	0.055
Qwen-7B+LoRA	0.014	0.012	0.033	0.029	0.101	0.092

TABLE 12. Per-system accuracy (3-seed mean, s.d.) vs majority vote vs oracle ensemble on the strict $w = 4$ test set ($n = 933$). The 0.061 gap between the best individual system (InternLM-7B+LoRA, 0.923) and the oracle bound (0.984) marks the headroom that an ideal ensemble could in principle exploit. Majority vote (0.923) ties the best individual, indicating that the systems are not yet diverse enough for naive voting to dominate.

Setting	Accuracy (3-seed mean, s.d.)
Individual: BERT-base	0.912 (0.006)
Individual: HateBERT	0.903 (0.002)
Individual: RoBERTa-base	0.902 (0.016)
Individual: DeBERTa-v3	0.915 (0.002)
Individual: InternLM-7B+LoRA	0.923 (0.002)
Individual: Qwen-7B+LoRA	0.918 (0.004)
Majority vote of 6 systems	0.923 (0.004)
Oracle (any-of-6 correct)	0.984 (0.003)

Under the routing policy of Eq. 9 every input pays the encoder pass, and the routed fraction additionally pays the LLM pass, so the expected latency of the executed cascade is

$$\mathbb{E}[\ell \mid \Pi_\tau] = \ell_{\text{enc}} + f(\tau) \cdot \ell_{\text{llm}}, \quad (10)$$

with $f(\tau)$ the fraction of traffic that the executed policy actually hands to the LLM. We execute the routing policy of Eq. 9 end-to-end at five thresholds $\tau \in \{0.60, 0.70, 0.80, 0.90, 0.95\}$: for every test sample the deployed prediction and probability come from the encoder when $\max(p_{\text{enc}}, 1 - p_{\text{enc}}) \geq \tau$ and from the LLM otherwise, with the two arms paired seed-by-seed across the three seeds, and every reported quantity (routed fraction, ECE,

Macro-F1, accuracy) is computed on the resulting mixed prediction stream. No accuracy, calibration, or routed-fraction operating point in Table 13 is interpolated; average latency is estimated via Eq. 10 from the measured endpoint latencies and the empirical routed fraction, with the encoder pass included for every input. The $f = 0$ and $f = 1$ rows coincide with the family means of Table 9 by construction.

The executed policy exposes three effects that a convex interpolation between the endpoints would not. First, the routed fraction is distribution-dependent: at $\tau = 0.80$ the policy hands 40.5% of OOD traffic to the LLM but only 9.9% of in-domain traffic, because the encoder’s confidence deficit under shift automatically diverts precisely the shifted inputs. The cost of a fixed threshold therefore adapts to the deployment distribution.

Second, calibration gains are real but smaller than a linear reading of the endpoints would suggest: executed $\tau = 0.90$ lowers the family-mean OOD ECE from 0.116 to 0.091 (a 22% relative reduction) at 260 ms, and $\tau = 0.95$ reaches 0.078 (33%) at 308 ms, still 33% below the 464 ms pure-LLM ceiling. The shortfall against the convex bound has a mechanistic reading consistent with the Calibration Cliff: the samples the encoder *keeps* are its high-confidence ones, and under shift these are exactly where encoder overconfidence concentrates (Section VI-E). Third, accuracy benefits arrive earlier than calibration benefits: OOD Macro-F1 rises from 0.630 to 0.698 already at $\tau = 0.80$ (190 ms), within 0.014 of the LLM-only 0.712, and in-domain accuracy at $\tau = 0.95$

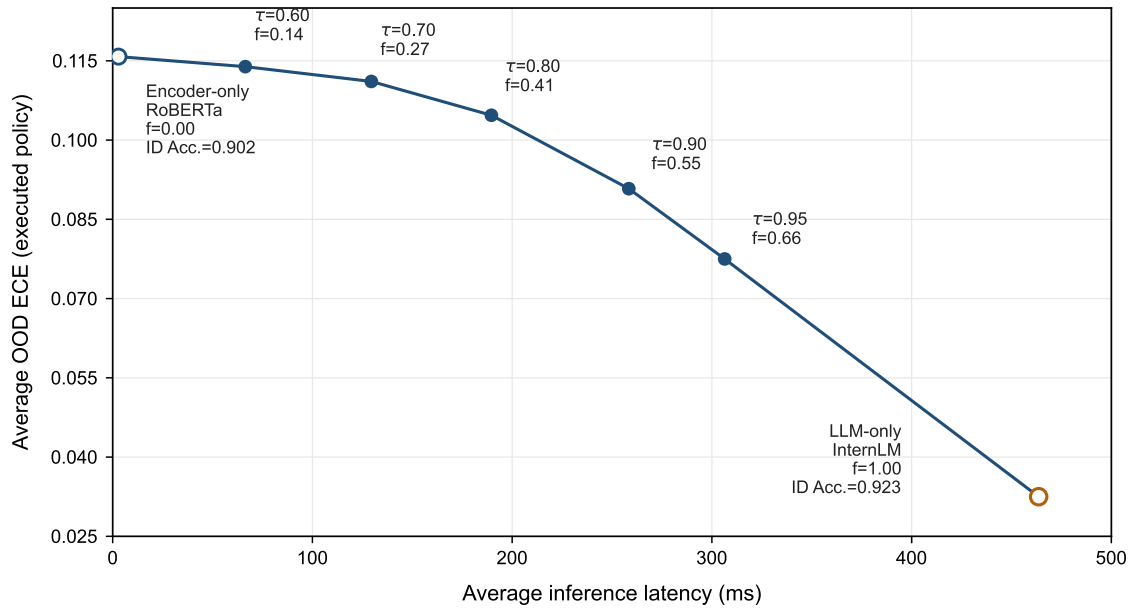


FIGURE 10. Latency-versus-calibration trade-off under the confidence-routing policy, executed end-to-end at every operating point (3 seeds, paired arms). Point labels give the threshold τ and the empirical OOD routed fraction f . In-domain accuracy of the executed policy rises from 0.902 (encoder-only) to 0.925 at $\tau = 0.95$ ($f_{ID} = 0.24$), versus 0.923 for LLM-only.

(0.925, at 23.5% LLM traffic) slightly exceeds the LLM-only endpoint (0.923). The oracle headroom of 0.061 in accuracy from Section VI-F remains an upper bound on what any router over the six-system panel can achieve, and the confidence-routing policy captures only the fraction for which the encoder's max-probability is informative about its own correctness; closing the residual is the ensemble-diversity problem [44], [57]. The recipe inherits the cascade template used in recent QA and reasoning systems [58], [59]: the confidence channel that selects “ask the LLM” is the same channel that the Calibration Cliff exposed in Section VI-E, and the compute footprint in Table 14 confirms that the trade-off is realised on commodity hardware, with a single RTX 4090 holding both the encoder and a 4-bit quantised LoRA-adapted 7B LLM.

H. COMPUTE AND ERRORS

1) COMPUTE FOOTPRINT

Because we frame the choice between encoder and LoRA-adapted 7B LLM as a deployment decision, Table 14 reports the compute and memory cost on a single NVIDIA RTX 4090 (24 GB), including total parameters, trainable parameters, peak inference VRAM, and median per-utterance latency on a fixed multi-turn prompt, to be read jointly with the in-domain and cross-domain accuracy tables as a cost-accuracy trade-off.

I. SYNTHETIC-TO-AUTHENTIC TRANSFER

The strict test set contains a single authentic conversation, so the submitted version of this work could not say whether its synthetic-to-authentic observations were representative.

We close this gap with a synthetic-only protocol: all ten authentic WhatsApp conversations (2,192 utterances) are removed from training and development and promoted to a held-out authentic evaluation suite, while the synthetic train/dev/test membership of the strict split is otherwise unchanged. All five systems are retrained from scratch under this protocol (3 seeds each, identical hyperparameters).

Table 15 reports the outcome, together with the external CGA benchmark (Conversations Gone Awry [60]: authentic Wikipedia talk-page conversations, official test split, 5,205 utterances, 8.1% personal-attack prior), which we evaluate with the original-protocol systems as a fourth out-of-domain target. Because its personal-attack label is adjacent to, rather than identical with, cyberbullying, CGA serves as an authentic-register adjacent-task check and not as a substitute for a same-task cyberbullying benchmark. The authentic-cluster suite covers five of the six systems (all four encoders and InternLM, each retrained with three seeds); on CGA we evaluate the two routing endpoints as a proof-of-concept external check, and extending the full panel to external authentic benchmarks is left to future work.

Three findings emerge. First, the synthetic-to-authentic Macro-F1 drop is large and family-independent: every system falls from ≈ 0.90 on the synthetic test set to 0.61–0.69 on the authentic suite, and the LLM's cross-domain F1 advantage vanishes under this register shift. Second, the single authentic conversation of the strict test set is unrepresentative in the direction the reviewer suspected: per-cluster Macro-F1 spans 0.48–0.87 across the ten conversations, and the original test conversation is the second-easiest of the ten. Third, the Calibration Cliff replicates in direction on both authentic

TABLE 13. Confidence-routing trade-off curve, executed end-to-end at every operating point (3 seeds, paired arms). $f_{\text{OOD}}(\tau)$ is the empirical routed fraction pooled over {CivCom, OLID, HateX}; OOD ECE and OOD Macro-F1 are computed on the mixed prediction stream of the executed policy; latency follows Eq. 10 with the empirical f_{OOD} (the encoder pass is paid on every input). $f_{\text{ID}}(\tau)$ and ID accuracy report the same executed policy on the in-domain strict $w = 4$ test set: the same threshold routes far less in-domain traffic than OOD traffic, so a single τ yields distribution-dependent cost.

Policy	$f_{\text{OOD}}(\tau)$	Avg. latency (ms)	OOD ECE	OOD Macro-F1	$f_{\text{ID}}(\tau)$	ID Acc.
Encoder-only (RoBERTa)	0.000	2.9	0.116	0.630	0.000	0.902
Routing ($\tau=0.60$)	0.138	66.7	0.114	0.660	0.031	0.908
Routing ($\tau=0.70$)	0.275	130.3	0.111	0.682	0.059	0.914
Routing ($\tau=0.80$)	0.405	190.8	0.105	0.698	0.099	0.918
Routing ($\tau=0.90$)	0.555	260.1	0.091	0.707	0.158	0.922
Routing ($\tau=0.95$)	0.659	308.3	0.078	0.710	0.235	0.925
LLM-only (InternLM)	1.000	463.6	0.032	0.712	1.000	0.923
Oracle bound on $f=f^*$	—	—	—	—	—	0.984

TABLE 14. Compute and memory footprint of every model family used in this paper, measured on a single RTX 4090. Latency is the median per-utterance forward (encoders) or 8-token greedy generation (LLMs) over 50 warm iterations on the same prompt; the LLM rows are 4-bit NF4 quantised.

Model	Params	Trainable	VRAM (MB)	Latency (ms)
BERT-base-uncased	109M	109M (100.00%)	429	2.7
RoBERTa-base	125M	125M (100.00%)	488	2.9
HateBERT	109M	109M (100.00%)	430	2.6
DeBERTa-v3-base	184M	184M (100.00%)	724	11.0
InternLM2.5-7B + LoRA	7.7B	18.9M (0.24%)	5,228	463.6
Qwen2.5-7B + LoRA	7.6B	20.2M (0.27%)	5,698	418.9

settings, at reduced magnitude: on the authentic suite the LLM’s ECE (0.120 ± 0.011) is below that of every encoder in all 12 matched-seed comparisons, and indeed in all 36 cross-seed pairings (worst LLM seed 0.131 vs. best encoder run 0.163; family mean 0.191, ratio $1.6\times$), and on CGA the ratio is $2.1\times$ (0.024 vs 0.052). Combined with the three platform-shift targets ($3.6\times$ – $7.8\times$, §VI-E2), the cliff’s direction is now consistent across five distribution-shift settings while its magnitude depends on the shift type: largest under platform shift, moderate under authentic register shift.

The executed routing policy of §VI-G marks the boundary of its own applicability here. On the authentic suite the encoder remains *confidently wrong*: at $\tau = 0.80$ the policy routes only 4.6% of authentic traffic (versus 40.5% under platform shift), so the mixed stream inherits the encoder’s miscalibration (ECE 0.195 at $\tau = 0.80$ vs 0.191 encoder-only) and recovers none of the LLM’s calibration advantage. Confidence routing helps exactly when the encoder’s confidence channel carries the shift signal; under register shift it does not, and closing this gap requires a learned reliability signal rather than the raw maximum probability (§VII).

J. QUALITATIVE ERROR ANALYSIS

Table 16 contrasts RoBERTa-base and InternLM2.5-7B + LoRA (both seed 42) on representative misclassifications from the strict- $w = 4$ test set, drawn from the slices analysed in Section VI-B3, plus one correctly classified positive control. We discuss five recurring error patterns and their apparent causes, then close with the out-of-domain failure mode that motivates the routing policy.

The first pattern is register-driven misses by the encoder. When harmful content is wrapped in a humorous or sarcastic register (e.g., “*I mean, come on. VCTM in tights? That’s a sight.*”), RoBERTa treats the dismissive tone as benign and predicts not_harmful at low harm probability (0.07), even though it handles the same kind of harm when phrased directly (last row). The LLM recovers all three register-driven rows of Table 16, but with moderate confidence (0.55–0.88) rather than certainty, consistent with its better-calibrated probability behaviour in Section VI-E.

The second pattern is implicit threats. “*Maybe, but [VCTM] better watch their back next time*” carries no lexical marker of abuse; RoBERTa scores it 0.25 while the LLM flags it at 0.56. The encoder’s miss is consistent with reliance on surface toxicity cues; the LLM’s thin margin suggests the pattern is hard rather than solved.

The third pattern is annotation-boundary cases around slurs-as-quotation. The fourth row repeats an earlier slur inside a conversation about that slur; the SynBullying annotation labels the repetition not_harmful (*is_hate_speech* = yes does not imply harm in this corpus, see §VI-B3), while both systems confidently predict harmful (0.93 and 0.85). We read such rows as label-boundary noise rather than model failure; they are the dominant source of the depressed Macro-F1 on the *is_hate_speech* = 1 slice.

The fourth pattern is LLM over-flagging of victim pushback and frustration. Of the 32 in-domain cases where the LLM errs while RoBERTa is correct (seed 42), most are not_harmful utterances in a charged context: victim meta-commentary (“*I can take a joke, but this feels more like ridicule*”, LLM 0.60), bystander frustration

TABLE 15. Synthetic-to-authentic transfer under the synthetic-only protocol (left block: no authentic data in train/dev; evaluation on all ten authentic WhatsApp conversation clusters; 3 seeds, mean $\pm$ s.d.) and the external CGA benchmark (right block: original-protocol systems on the official CGA test split). ECE is the 15-bin estimator of Eq. 4. Cluster-bootstrap 95% CIs for RoBERTa authentic Macro-F1 span [0.60, 0.76] across seeds ($B = 10,000$ over the ten clusters).

Model	Syn. test F1	Synthetic-only protocol			CGA (original protocol)	
		Auth. F1	Auth. harm-F1	Auth. ECE	F1	ECE
BERT-base	0.893	0.630	0.404	0.198	—	—
HateBERT	0.901	0.608	0.361	0.204	—	—
DeBERTa-v3	0.904	0.694	0.521	0.173	—	—
RoBERTa-base	0.918	0.677	0.490	0.191	0.664	0.052
InternLM2.5-7B+LoRA	0.908	0.675	0.482	0.120	0.667	0.024

(“Whatever, I’m done with [VCTM]. Can’t stand snitches”, LLM 0.64), and harsh-but-annotated-benign banter. The LLM appears to score the conversational atmosphere where the annotation scores the target utterance’s own act. The reverse pool (RoBERTa errs, LLM correct) is larger (40 cases) and concentrated in the first two patterns.

The fifth pattern concerns the authentic WhatsApp slice, where both systems degrade for a distinct reason: pragmatic ambiguity in noisy, real adolescent talk. In-group profanity between friends (“Bitch!”, annotated not_harmful in context) is confidently flagged by both systems (0.97/0.96); elliptical utterances (“She could not stay silent then”, annotated harmful) hide the harm in discourse context that neither system resolves; and defensive capitals (“I’M NOT HIS GIRLFRIEND.”) draw an encoder false positive (0.68). Unlike the synthetic-slice patterns, these errors are less about register and more about interactional history, which supports the deeper synthetic-data limitations discussed in Section VII.

Finally, the out-of-domain slices exhibit the confidently-wrong failure mode that the Calibration Cliff quantifies: across OLID, HateXplain, and Civil Comments (seed 42), 156 utterances are misclassified by RoBERTa at confidence ≥ 0.9 while the LLM is correct, typically benign posts with dense surface toxicity cues, such as emoji-laden promotional posts flagged as harmful at 0.94–0.96, alongside overtly hateful content scored as benign at 0.05–0.08. These are precisely the inputs a confidence router cannot save, because the encoder never asks for help on them, which is why the executed routing curve in Section VI-G improves calibration more slowly than accuracy.

VII. DISCUSSION

A. THE CALIBRATION CLIFF

The audit’s main finding is sharper than an F1-only analysis would suggest. In domain, no system separates from the cluster once the resampling unit matches the data-generating unit: each of the five Holm-corrected contrasts against RoBERTa-base on SynBullying Strict-w = 4 comes out at $p_{\text{Holm}} = 1.00$ under the conversation-level cluster bootstrap, and an applied team would struggle to justify the $60\times$ parameter overhead of the 7B LLM on in-domain F1 alone. Out of domain, the same overhead buys a substantive Macro-F1 gain (+0.054 on OLID, +0.148 on HateXplain,

and +0.075 on Civil Comments at s.d. 0.066) and a more consequential between-panel gain in Expected Calibration Error. The Calibration Cliff reaches $14.9\times$ worst-vs-best on Civil Comments and persists at $9.4\times$ residual after temperature scaling, so a moderation pipeline consuming BERT-family probabilities under shift operates against a confidence channel roughly an order of magnitude noisier than the LLM panel’s. We therefore read this paper as recasting the cyberbullying-detection question from “which system has the highest in-domain F1?” to “which system’s confidence remains decision-grade under the distribution actually encountered at deployment time?”.

Three non-exclusive mechanisms are consistent with the LLM advantage under shift. The encoder baselines were pre-trained on relatively homogeneous corpora, whereas the LoRA-adapted 7B chat models saw orders of magnitude more diverse text and have encountered surface forms that the encoders have not. The LoRA adapter modifies only 0.24%–0.27% of the base-model parameters, so the bulk of the model retains its broad instruction-following prior on inputs whose surface diverges from the training set, whereas a fully fine-tuned encoder has overwritten its general-purpose representations with task-specific decision boundaries. Generative framing further inherits more of the pre-training distribution’s invariances than a softmax classifier head, as the ≤ 0.0013 Macro-F1 agreement between free-form, constrained, and verbalizer-scoring decoding (Section VI-D1) confirms. These mechanisms are not separated by an experiment here; the joint accuracy and calibration evidence is consistent with all three.

A fourth, non-mechanistic confound deserves explicit acknowledgement. The training distribution is the SynBullying corpus, which is itself synthesised by three large language models (GPT-4o, Grok-2, LLaMA-3.3) plus a small authentic WhatsApp slice, while the three out-of-domain targets (OLID, HateXplain, Civil Comments) are drawn from real social-media and news platforms. The LoRA-adapted LLM arm shares pretraining-distribution overlap with one of these surfaces (the synthetic in-domain text) that the encoder arm does not, so part of the cross-domain advantage we observe could in principle reflect distributional proximity rather than a property of the encoder-versus-LLM

TABLE 16. Selected qualitative predictions of RoBERTa-base and InternLM2.5-7B + LoRA (both seed 42) on the strict- $w = 4$ test set. The first five rows are misclassifications representative of the linguistic-marker slices analysed in §VI-B3; the last row is a correctly classified positive control. Cells give the predicted label with the predicted harm probability in parentheses; wrong predictions are marked †. “CB type” is the SynBullying CB_types field.

Utterance (target only; window omitted)	CB type	True	RoBERTa	InternLM	Flag / Pattern
“Bet you love all the extra homework, right VCTM? More time to spend alone with your books.”	Insult_General	harm	not (0.43) [†]	harm (0.80)	is_sarcastic; sarcastic taunt
“I mean, come on. VCTM in tights? That’s a sight.”	Body_Shame	harm	not (0.07) [†]	harm (0.55)	is_humorous; joke-framed body shaming
“Passionate about dance? You mean passionate about wearing makeup and looking like a fag?”	Sexism	harm	not (0.50) [†]	harm (0.88)	is_hate_speech; overt slur
“Oh, so now you’re saying you’re better than us? You’re just a pathetic little ballet boy who thinks he’s above everyone else.”	Sexism	not	harm (0.93) [†]	harm (0.85) [†]	is_hate_speech; slur repetition, label boundary
“Maybe, but [VCTM] better watch their back next time.”	Threat	harm	not (0.25) [†]	harm (0.56)	implicit threat; subtle GPT-4o style
“Haha, dude, ballet is for girls. What are you doing dancing around in tights?”	Sexism	harm	harm (0.91)	harm (0.86)	is_humorous; control (both correct)

confidence channel. The argument that this confound does not exhaust the observed effect is twofold. First, the same encoder-versus-LLM gap appears across all three OOD targets simultaneously and at heterogeneous class priors (8%, 28%, 59% harmful); a pure distributional-proximity account would predict heterogeneous gap direction across these priors. Second, the family-mean Brier under HateX-plain (whose language is closest to the encoders’ Reddit pretraining for HateBERT) still shows the same direction of advantage to the LLM panel. We nevertheless flag a hold-out on a fully authentic in-domain cyberbullying corpus as the cleanest follow-up that would separate this confound from the calibration mechanism.

B. SYNTHETIC-DATA LIMITATIONS

Beyond the panel-comparison confound above, training and testing on LLM-synthesised cyberbullying carries limitations that bound what any study on this corpus, including ours, can claim. First, generator-style artefacts: the synthetic conversations are produced by three generators following four scripted scenario templates, so harmful acts arrive at a scripted density and in fluent, low-noise prose. Authentic adolescent talk in the WhatsApp slice is elliptical, misspelled, emoji-laden, and pragmatically ambiguous, and our error analysis (§VI-J) shows that both model families fail there for reasons, such as interactional history and in-group profanity, that the synthetic slices barely exercise. Second, construct narrowing: with four scenario templates, the topical and relational diversity of real cyberbullying (persistence over days, cross-platform pile-ons, image-based abuse) is not represented; a detector performing well here may not transfer to those modalities. Third, an evaluation circularity risk specific to LLM-generated benchmarks: models from the same family as a generator may be advantaged on generator-authored test text, and annotation pipelines that use an LLM

in the loop can import the generator’s notion of harm into the gold standard. Our per-source slicing (GPT-4o text is hardest for every system, §VI-B2) shows the generators are not interchangeable, which mitigates but does not eliminate this risk.

For these reasons we treat headline in-domain numbers as protocol-bounded quantities, and we anchor the deployment-facing claims on the authentic and out-of-domain evidence instead. The synthetic-only study of §VI-I quantifies the cost directly: without authentic conversations in training, every system loses 0.21–0.29 Macro-F1 between the synthetic test set and the ten authentic clusters, and the drop is as large for the LoRA-adapted LLM as for the encoders. Synthetic corpora of this kind support protocol research and relative audits; absolute performance claims for deployment require authentic evaluation data.

C. CONFIDENCE ROUTING

The Calibration Cliff identifies the failure mode; the routing analysis of Section VI-G converts that failure mode into a deployable recipe, now executed end-to-end at every reported operating point. Executed $\tau = 0.90$ delivers a 22% relative reduction in family-mean OOD Expected Calibration Error at 260 ms average latency on a single RTX 4090, 44% below the pure-LLM cost, and OOD Macro-F1 recovers most of the LLM’s cross-domain advantage already at $\tau = 0.80$. Two executed-policy properties matter for deployment beyond the curve itself. The routed fraction is distribution-dependent: the same threshold spends roughly four times more LLM budget on shifted traffic than on in-domain traffic, so capacity planning should treat f as a function of the deployment distribution rather than a constant. And because the encoder retains exactly its high-confidence slice, where OOD overconfidence concentrates, executed calibration improves more slowly than a convex reading of the

two endpoints would suggest. A deployment team can read the operating point off Table 13 directly given its production latency budget and its tolerance for residual high-confidence error.

The routing analysis is an initial component rather than a complete reliability solution. The oracle headroom of Section VI-F shows that an ideal router over the six-system panel could reach accuracy 0.984, 0.061 above the best individual system. Closing the rest of that gap is the ensemble-diversity problem, and the disagreement pattern (every BERT cell worse than every LLM cell on every OOD target in Table 9) hints at how to attack it. Distance-aware calibration [56], isotonic regression on logits, conformal abstention [58], [59], and LoRA-tuned BERT probes are the natural follow-ups; whether any of them closes the residual portion of the Cliff at the encoder-side latency budget remains an open empirical question for future work.

We emphasise that the routing recipe is an evaluated component, not a deployment-ready moderation system. Production moderation embeds any classifier in a wider socio-technical pipeline whose constraints our benchmark does not measure: human review capacity and appeal processes for contested decisions, per-jurisdiction privacy obligations when adolescent conversations are processed (including data-minimisation and retention limits that may preclude sending full context windows to a large model), tail latency and batching behaviour under production load rather than the batch-1 median we report, fairness auditing across user subgroups, and the ethical requirement that automated confidence scores support rather than replace human judgement on borderline harms [61]. Our latency-versus-calibration curve should therefore be read as evidence that a favourable trade-off region exists on commodity hardware, and as an input to, not a substitute for, the system-level and ethical engineering that moderation deployment requires.

D. ENCODER COMPETITIVENESS

The most actionable in-domain finding is empirical: under strictly leakage-controlled evaluation, all four compact encoders and both LoRA-adapted 7B LLMs cluster in a single Macro-F1 band around 0.91; none of the five Holm-corrected contrasts against RoBERTa-base separates from it under the cluster bootstrap. This cuts against recent work [6], [7] that treats LLM adaptation as the default for cyberbullying detection. On a single-utterance binary task the linguistic signal that a properly fine-tuned encoder can capture is already saturating the distribution, and LoRA-only adaptation modifies too few parameters to add a separable margin. For deployment, an applied system that needs in-domain detection on similar conversational data should default to a strong encoder under the routing policy of Section VI-G, escalating to the LLM only on the fraction of inputs whose confidence falls below threshold.

E. CONTEXT AND CROSS-SOURCE

Two further patterns concern context window and source slicing. Section VI-B1 shows that the window-quality curve is monotonically non-decreasing up to $w = 8$ once the positional budget is matched, and that a naive run with a fixed 256-token cap on $w = 8$ produces an apparent inverted-U that is an artefact of right-side truncation removing the target utterance, not a property of context; $w \in [4, 8]$ with a ≥ 512 -token budget is therefore a defensible default. Section VI-B2 shows that the inter-generator axis matters more than the synthetic-versus-authentic axis: GPT-4o-generated data is the hardest source (strongest encoder reaches only 0.606 per-class harm-F1), and the authentic WhatsApp conversation sits among the easier sources. The two LLM systems modestly improve on the GPT-4o slice (Qwen2.5-7B reaches 0.828 Macro-F1, the highest of any system); the authentic-data finding is preliminary and constrained by a single conversation.

F. PLATEAU AND FAILURES

The in-domain band is not a task ceiling: 82% of test utterances are answered identically by all six systems, and the remaining 18% elicit disagreement on which an oracle that takes any-of-six correct reaches 0.984 overall accuracy, a 0.061 headroom over the best individual (0.923). The plateau should therefore not be read as evidence of a corpus-level performance ceiling, and a routing hybrid that assigns disagreement-region utterances to the LLM, formalised as the policy of Eq. 9, recovers a substantive fraction of that gap. The remaining errors fall into three patterns that the slice analysis (Section VI-B3) and the qualitative examples (Table 16) make concrete. Sarcasm and humour-framed insults are missed at roughly twice the overall miss rate, a classically known failure mode of toxicity classifiers [5], because the wrapper deliberately suppresses the surface lexical signal. Hate-speech-flagged utterances are sometimes labelled not_harmful in SynBullying because they appear as quotations or supportive call-outs, and our classifier still treats them as harmful; whether this is label error or substantive annotator disagreement is a downstream question we report transparently rather than resolve. Implicit threats that read as innocuous in isolation but become threatening in conversational context are the residual false negatives that the $w = 4$ window is not always sufficient to catch, and an explicit conversational intent model is the natural next step.

VIII. CONCLUSION

This work benchmarks four compact transformer encoders against two LoRA-adapted 7B large language models on the SynBullying corpus, under a conversation-level evaluation protocol that combines speaker anonymisation, three-seed averaging, conversation-cluster paired bootstrap with Holm–Bonferroni correction, and a five-fold Group-KFold robustness check. Six otherwise different systems collapse into a single in-domain Macro-F1 band near 0.91,

while a between-panel discontinuity emerges under cross-distribution shift: encoder-panel Expected Calibration Error reaches up to $14.9\times$ the LoRA-adapted LLM-panel floor on Civil Comments and persists at $9.4\times$ residual after temperature scaling. We name this discontinuity the Calibration Cliff, and we show, by executing the routing policy end-to-end at five thresholds, that a lightweight confidence-routing policy reduces mean out-of-domain Expected Calibration Error by 22% at $\tau = 0.90$ at 44% below the pure-LLM latency on commodity hardware, while adaptively routing more traffic to the LLM under shift than in-domain.

As automated moderation pipelines increasingly consume probabilities rather than labels, the dominant evaluation signal in cyberbullying-detection literature, in-domain F1, becomes insufficient to characterise deployment behaviour. Our findings support a narrower claim: for this dataset, label mapping, and model panel, the practical value of LLM adaptation lies more in calibration and cross-distribution robustness than in in-domain headroom, and the executed routing policy provides an evaluated encoder/LLM trade-off on commodity hardware. Because confidence reliability under shift bears directly on the welfare of users that moderation systems are designed to protect, joint accuracy-and-calibration auditing should become the default rather than an afterthought in future cyberbullying-detection studies.

A. LIMITATIONS AND FUTURE WORK

This study has several limitations. First, the Civil Comments cross-domain delta of $+0.075$ Macro-F1 carries a standard deviation of 0.066 across the three RoBERTa source seeds (paired per-seed deltas $+0.131$, $+0.092$, $+0.002$), reflecting the extreme 8%-harmful class prior; we therefore treat OLID and HateXplain (s.d. 0.003 and 0.007) as the primary cross-domain F1 evidence and Civil Comments as a calibration stress test. Second, the strict protocol anonymises every speaker prefix, but the SynBullying corpus embeds the role markers VCTM and BULLY inside roughly 75% of training utterances, so headline Macro-F1 values should be read as an upper bound. Third, although the synthetic-only study of §VI-I now evaluates all ten authentic conversations of the corpus and adds an external authentic benchmark, the authentic evidence still derives from role-play data and a single platform register per source; organic cyberbullying traces remain the missing gold standard. The authentic suite also comprises only ten conversation clusters, so its cluster-bootstrap confidence intervals are correspondingly wide, and the authentic-domain contrasts should be read at that resolution rather than as tight point estimates. Fourth, all in-domain numbers come from a single conversation-level split with three training seeds; the 5-fold GroupKFold check captures across-split variance, but a leave-one-conversation-out protocol on a larger authentic hold-out remains a natural extension.

Fifth, the panel is asymmetric (four encoders against two LoRA-adapted LLMs), so family-mean contrasts average

TABLE 17. Symbols used throughout the paper.

Symbol	Meaning
$\mathcal{C}, N$	set of conversations, $N=130$
c_n, T_n	conversation n with T_n utterances
$u_t^{(n)}, s_t^{(n)}$	utterance t and its speaker in c_n
$y_t^{(n)} \in \{0, 1\}$	binary harm label (0=not_harmful, 1=harmful)
$w \in \{0, 1, 2, 4, 8\}$	context window size (#prior utterances)
$x_w(t, n)$	speaker-prefixed window input to the model
$\pi(s)$	speaker anonymisation map $\pi: \mathcal{S}_n \rightarrow \mathbb{N}$
$\mathcal{D}_{tr}, \mathcal{D}_{dev}, \mathcal{D}_{te}$	conversation-level train/dev/test partitions
f_θ	encoder classifier with parameters θ
$\Delta\theta_{LoRA}$	low-rank LoRA update
r, α	LoRA rank and scaling
$\hat{y}_t$	predicted harm label
F_1^{macro}	macro-averaged F1 score
B	number of bootstrap resamples

TABLE 18. Composition of the strict test set by source.

Source	#Utt.	%Harmful
GPT-4o	319	8.5%
Grok-2	350	64.9%
LLaMA-3.3	143	18.9%
WhatsApp (authentic)	121	34.7%
Total	933	34.6%

over unequal group sizes; the LoRA-rank ablation is single-seed, and OLID's OFFENSIVE label is an approximation of HARMFUL rather than its synonym, dampening the OLID Cliff of $4.88\times$ raw; we report Civil Comments and HateXplain as headline ratios, in each case under the evaluated configuration. A hierarchical-encoder baseline that aggregates per-utterance [CLS] tokens trails the flat baseline by 0.014 Macro-F1: under the evaluated implementation, the $\sim 14M$ -parameter aggregator brings no accuracy benefit on short ~ 5 -utterance windows. Finally, the audit is validated under English-language Western-internet harms; cross-cultural out-of-distribution validation on HateDay [20] and on non-Western corpora is required before the Calibration Cliff is treated as universal.

Future work along three directions builds directly on this study. Distance-aware calibration [56] and isotonic regression on the encoder logits may close more of the Calibration Cliff at the encoder-side latency budget. Calibration-aware routing that replaces the encoder's raw maximum probability with a learned reliability score is the natural way to capture the residual oracle headroom, with conformal abstention [58], [59] providing a principled framework. Cross-cultural out-of-distribution validation on HateDay [20] and on non-Western cyberbullying corpora is the prerequisite for treating the Calibration Cliff as a universal property of the BERT-family confidence channel rather than of the English-language Western-internet distribution under which it was measured. Beyond these, the SynBullying CB_types taxonomy supports a fine-grained classifier under the same conversation-level protocol,

TABLE 19. Dataset statistics for the SynBullying corpus, its strict partitions, and the OLID/OffensEval-2019 test split used for cross-domain transfer. Percentages denote the fraction of harmful utterances or active linguistic flags; utterance length is in white-space tokens.

Partition	#utt	#conv	%H	Utt. length (tok)			Linguistic flag (%)			Source mix (#utt)		
				avg	med	max	%sar	%hum	%hate	GPT-4o	Grok-2	LLaMA-3.3
SynBullying (all)	14,222	130	27.6	9.6	9	68	1.6	4.1	2.0	4,770	3,960	3,300
Strict train	11,681	104	28.4	9.4	9	68	1.8	4.1	2.1	4,011	3,410	2,603
Strict dev	1,608	13	17.8	9.5	9	43	0.6	4.0	0.3	440	200	554
Strict test	933	13	34.6	11.7	11	37	1.4	3.8	3.3	319	350	143
WhatsApp authentic slice	121	1	34.7	8.7	6	37	0.0	7.4	0.0	–	–	–
OLID test [33], [48]	860	–	27.9	15.4	13	79	–	–	–	(Twitter, single-tweet)		

and intervention-generation evaluation extends the audit from detection to response.

APPENDIX A EXPERIMENTAL SETUP DETAILS

A. HARDWARE

All experiments were conducted on a single workstation running Ubuntu 22.04, equipped with one NVIDIA RTX 4090D GPU (24 GB GDDR6X), an Intel Core i9 processor, 128 GB DDR4 memory, and a 2 TB NVMe SSD for dataset and checkpoint storage. The 4-bit NF4-quantised seven-billion-parameter language models fit comfortably in 24 GB of GPU memory with 5.7 GB headroom.

B. SOFTWARE

Models were implemented in Python 3.10 using PyTorch 2.4.1 with CUDA 12.1, HuggingFace Transformers 4.45.2, PEFT 0.12.0 for low-rank adaptation, BitsAndBytes 0.44.1 for 4-bit quantisation, and Accelerate 0.34.2 for mixed-precision training. Statistical analyses used NumPy 1.26.4 and SciPy 1.15.3. The cluster-bootstrap and Holm–Bonferroni implementations are released alongside the paper. Random seeds {41, 42, 43} were applied uniformly to PyTorch, NumPy, and Python’s built-in random module.

C. TRAINING TIME

Each compact encoder seed converges in approximately two minutes on the strict- $w = 4$ training partition; one epoch of LoRA fine-tuning on a seven-billion-parameter chat model takes about forty minutes. The full six-system three-seed sweep plus the cross-domain inference passes complete in under twelve GPU-hours.

APPENDIX B SUPPLEMENTARY TABLES

This appendix collects three tables that support but are not central to the main argument: the symbol glossary (Table 17), the test-set composition by source (Table 18), and the per-split dataset statistics (Table 19). Additional supplementary tables covering equivalence testing, the inference-strategy comparison, and the SynBullying Table 9 cross-reference are provided in the accompanying supplementary material.

APPENDIX C

NOTATION AND FORMALISM DETAILS

This appendix collects the auxiliary formal definitions that support Section III-F.

A. SPEAKER ANONYMIZATION

The raw SynBullying release labels speakers with role-revealing tokens (BULLY1, VCTM1, VSUP) that are near-perfectly correlated with y_t , a label-leakage pattern. We replace every speaker by a within-conversation index using the bijection

$$\pi(s_t^{(n)}) = k \quad \text{where} \quad k = |\{s_{t'}^{(n)} \in \mathcal{S}_n : t' \leq t\}|, \quad (11)$$

so the first speaker to appear becomes *Speaker1*, the second *Speaker2*, and so on. Anonymisation is applied after the strict split of Eq. 1.

B. CONTEXT WINDOW

For a target utterance $u_t^{(n)}$ with anonymised speakers, we construct the window of length $w + 1$ by concatenating the target with up to w preceding utterances:

$$x_w(t, n) = \bigoplus_{i=\max(1, t-w)}^t [\tau(\pi(s_i^{(n)})) \parallel u_i^{(n)}], \quad (12)$$

where $\parallel$ is string concatenation, $\bigoplus$ joins per-utterance strings with newline separators, and $\tau(k) = \text{“Speaker}k\text{:”}$ produces the speaker prefix. We left-truncate to a fixed token budget; the target utterance is always preserved.

C. MACRO-F1

Let $F_1^c(\mathcal{X})$ denote the class- c F1 on a subset $\mathcal{X}$. The Macro-F1 we report is the unweighted mean of the two class F1s:

$$F_1^{\text{macro}}(\mathcal{X}) = \frac{1}{2} [F_1^0(\mathcal{X}) + F_1^1(\mathcal{X})]. \quad (13)$$

D. CROSS-DOMAIN TRANSFER GAP

For a model trained on SynBullying strict- $w = 4$, we compute the cross-domain gap

$$\Delta^{\text{xd}}(M) = F_{1,M}^{\text{macro}}(\text{OLID}) - F_{1,M}^{\text{macro}}(\text{SynB strict-}w = 4), \quad (14)$$

which is negative for every system (in-domain is easier) but whose magnitude differs across panels. That magnitude difference is the empirical centre of Section VI-C.

E. ENCODER AND LLM LOSS

Encoder branches minimise the standard class-weighted cross-entropy with weights $\omega_c \propto 1/\Pr[y = c]$ estimated on $\mathcal{D}_{tr}$, yielding a harm-class weight $\omega_1 \approx 1/0.28$. The LoRA-adapted decoder LLMs use the standard negative log-likelihood on the gold answer tokens conditioned on the chat-template prompt. Per-source generalisation slices report the conditional Macro-F1 $F_{1,M}^{\text{macro}}(\mathcal{D}_{te}^g)$ on each source $g \in \{\text{GPT-4o, Grok, LLaMA, WhatsApp}\}$.

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ETHICAL CONSIDERATIONS

Cyberbullying detection is dual-use: false negatives expose victims to unchecked harassment, false positives chill legitimate participation, and over-action falls disproportionately on adolescents and racialized users. The reported ~ 0.91 Macro-F1 should not be read as deployment-ready, because training is largely on synthetic data, per-class F1 collapses on sarcasm, humour, and overt hate, and all in-domain numbers come from a single conversation-level split; any deployed system requires human-in-the-loop review, domain-shift monitoring, and demographic-fairness auditing [2]. The Calibration Cliff has a welfare implication that an F1-only evaluation does not capture: an over-confident encoder under shift translates directly into high-confidence false-positive moderation at exactly the operating points where automated action is most likely to be irreversible. The confidence-routing policy of §VI-G can reduce this risk, but it does not replace human review or deployment-specific safety evaluation. SynBullying is used under its original release terms; no human-subjects re-collection was performed. The released code, strict-split manifest, and per-example prediction artefacts are available under the MIT License; the trained LoRA adapter weights will be released upon publication under a license compatible with the base-model terms, together with documented intended-use guidance.

DATA AVAILABILITY

The SynBullying corpus is available from its original authors under the release terms; OLID is available via the `tweet_eval` subset [48]; HateXplain and Civil Comments are available from their respective public releases. The strict conversation-level split manifest, per-example test predictions, and the SHA-256-pinned summary artefact are publicly available at <https://github.com/liutinglei/liutinglei-CARE-Calibration-Cliff> under the MIT License. The trained LoRA adapter weights will be added to the same repository upon publication.

CODE AVAILABILITY

The full Python codebase, covering data preparation, training, evaluation, cluster bootstrap with Holm–Bonferroni correction, 15-bin ECE and Brier calibration analysis, the confidence-routing simulator, and figure generation, is publicly available at <https://github.com/liutinglei/liutinglei-CARE-Calibration-Cliff> under the MIT License. A frozen `environment.yml` will be added to the same repository upon publication.

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